

CEG 402: Introduction & Basic Concepts

Reading Assignment

- Pages 1-59

Introduction

Outline:

- What is a communication network
- Network taxonomy
- What is a protocol?
- Internet - example
- Protocol layers – Open System Interconnection OSI model
- Performance: loss, delay

What is a communication network?

- A set of equipment, facilities, and software that provides a service
 - That transfers information between users at various geographical locations

Communication Networks

- Examples
 - Telephone networks
 - Television broadcast networks
 - Cable networks
 - Satellite networks
 - Cellular networks
 - Computer networks
 - Home networks
 - The Internet

What is a Communication Network?

- Network offers a service:
 - move information
 - bird, fire, messenger, truck, telegraph, telephone, Internet ...
- What distinguishes different types of networks?
 - many
 - the services they provide

What services?

- Examples:
 - Distributed services
 - Client-server service:
 - ! voice, video, Web, e-mail, telnet, ftp, games, e-commerce, database
 - Peer-to-peer service:
 - ! file sharing, distributed storage

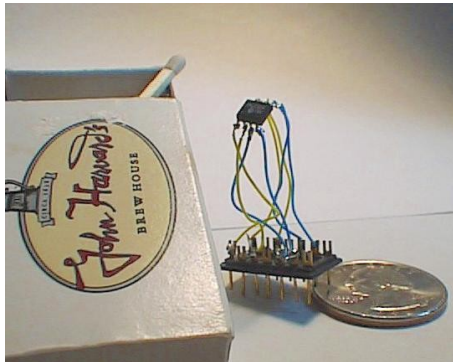
Internet appliances



IP picture frame
<http://www.ceiva.com/>



Sensors
<http://www.xbow.com>



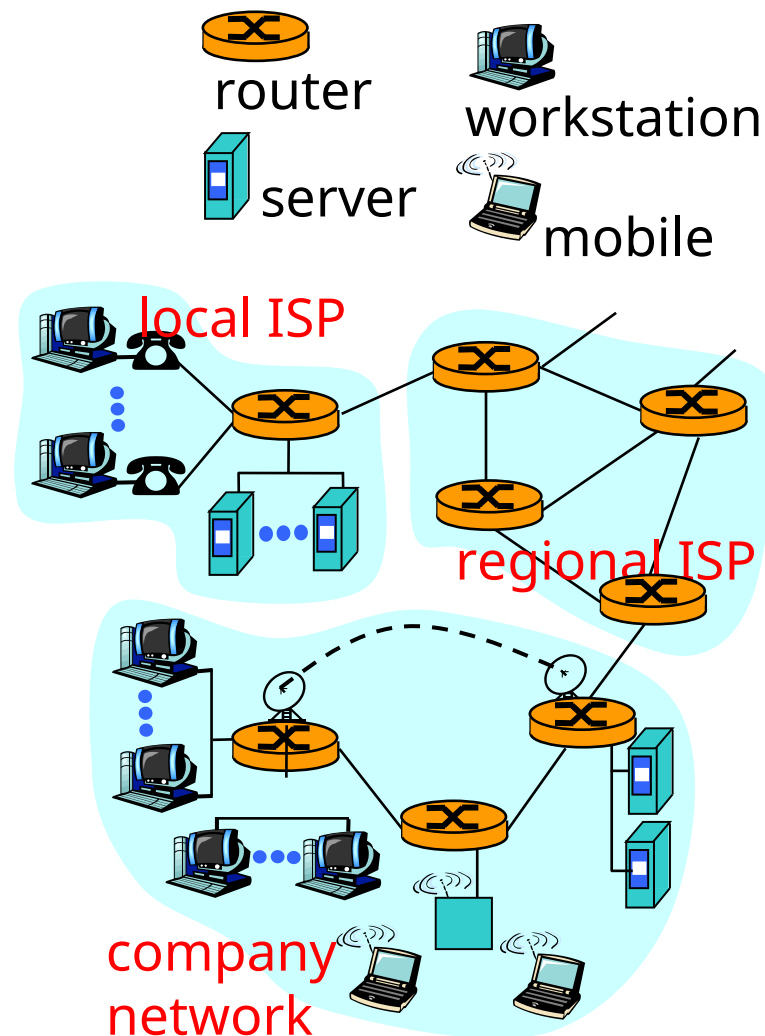
World's smallest web server
<http://www-ccs.cs.umass.edu/~shri/iPic.html>



Web-enabled toaster

What is a communication network?

- millions of connected computing devices: *hosts, end-systems*
 - PCs, workstations, servers
 - PDAs, phones, toastersrunning *network applications*
- *communication links*
 - fiber, copper, radio, satellite
 - transmission rate = *bandwidth*
 - *56kbps (dial-up)*
 - *1.5Mbps (DSL)*
- *routers*: forward packets (chunks of data)



Types of Networks

- Geographical distance
 - Local Area Networks (LAN): Ethernet, wireless LAN, Token ring, FDDI
 - Metropolitan Area Networks (MAN): Optical ring
 - Wide Area Networks (WAN): X.25, frame relay, ATM
- Information type
 - Data networks vs. telecommunication networks (voice)
- Application type
 - Special purpose networks:
 - airline reservation network, banking network, credit card network
 - General purpose network:
 - Internet

Types of Networks

- Right to use
 - private: enterprise networks
 - public: telephone network, Internet

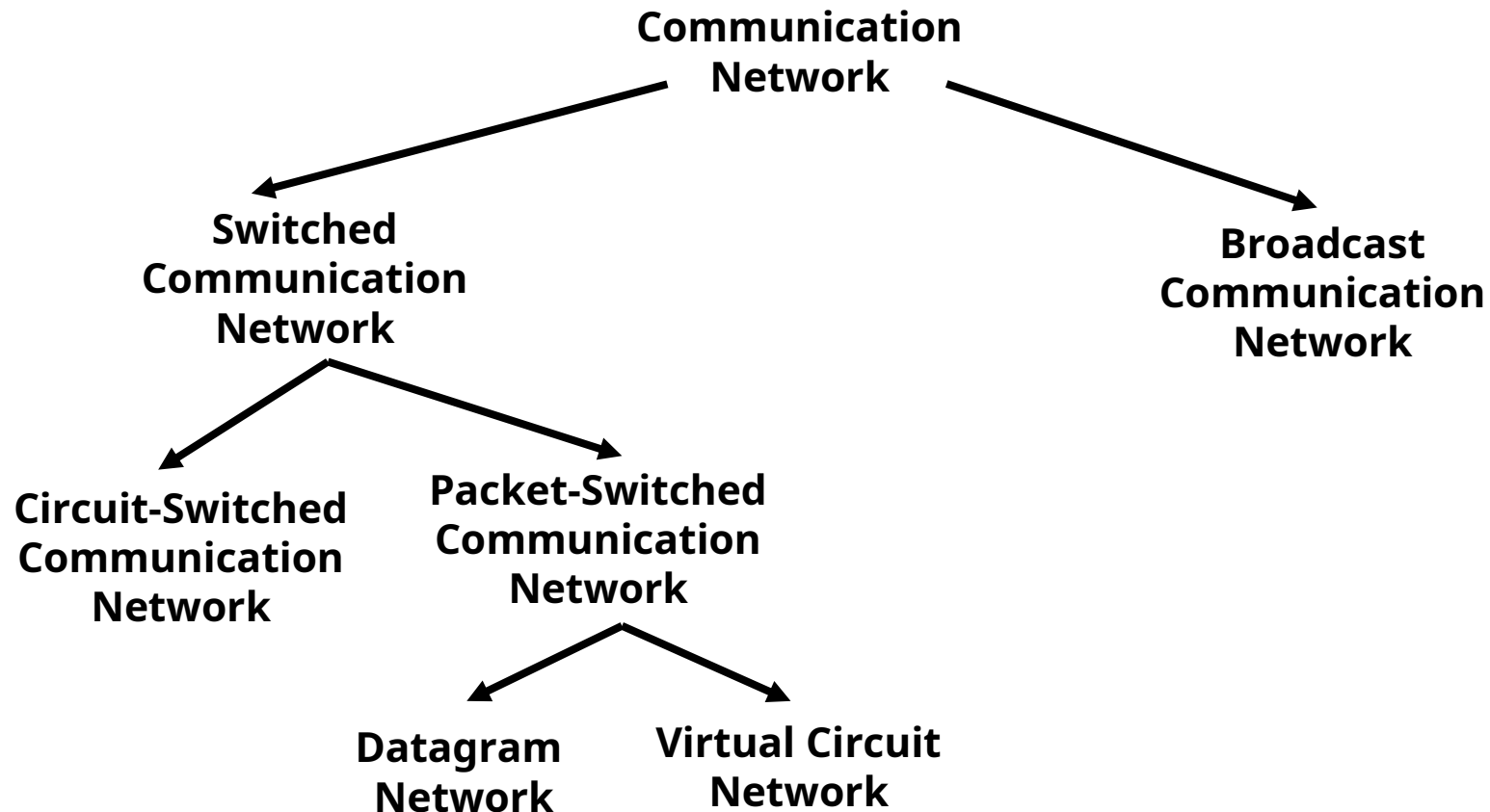
- Ownership of protocols
 - proprietary: voice over IP
 - open: Internet Protocol suite <http://www.ietf.org/>

The Internet

- Global scale, general purpose, heterogeneous-technologies, public, network
- Internet Protocols
 - open standard:
 - Internet Engineering Task Force (IETF) as standard body <http://www.ietf.org/>
 - technical basis for other types of networks
 - Intranet: enterprise IP network
- Developed by the research community

A Taxonomy of Communication Networks

- Communication networks can be classified based on the way in which the nodes exchange information:



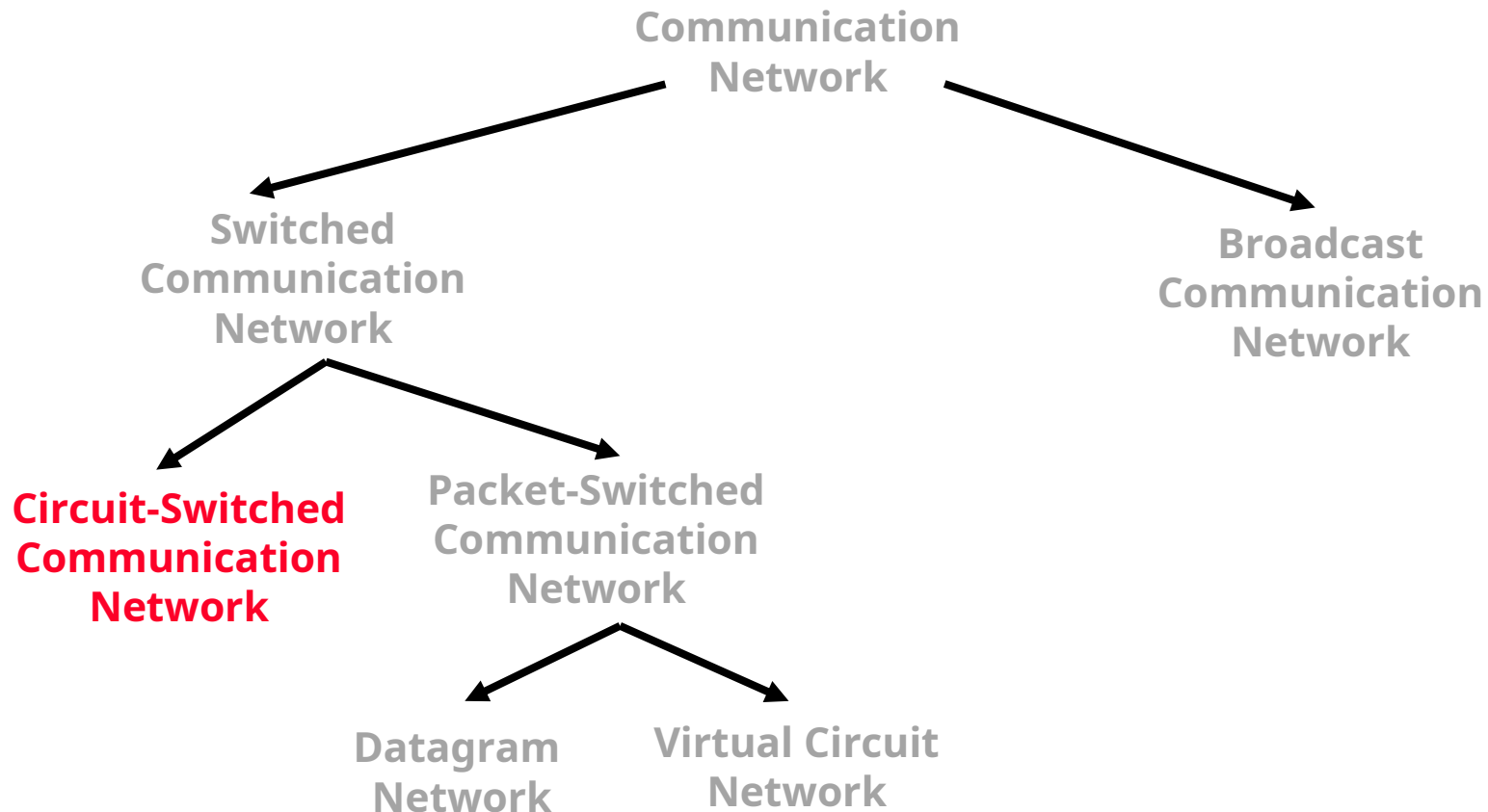
Broadcast vs. Switched Communication Networks

- Broadcast communication networks
 - information transmitted by any node is received by **every** other node in the network
 - examples: usually in LANs (Ethernet, wireless LAN)
 - Problem: coordinate the access of all nodes to the shared communication medium (Multiple Access Problem)

- Switched communication networks
 - information is transmitted to a sub-set of designated nodes
 - examples: WANs (Telephone Network, Internet)
 - Problem: how to forward information to intended node(s)
 - this is done by special nodes (e.g., routers, switches) running routing protocols

A Taxonomy of Communication Networks

- Communication networks can be classified based on the way in which the nodes exchange information:

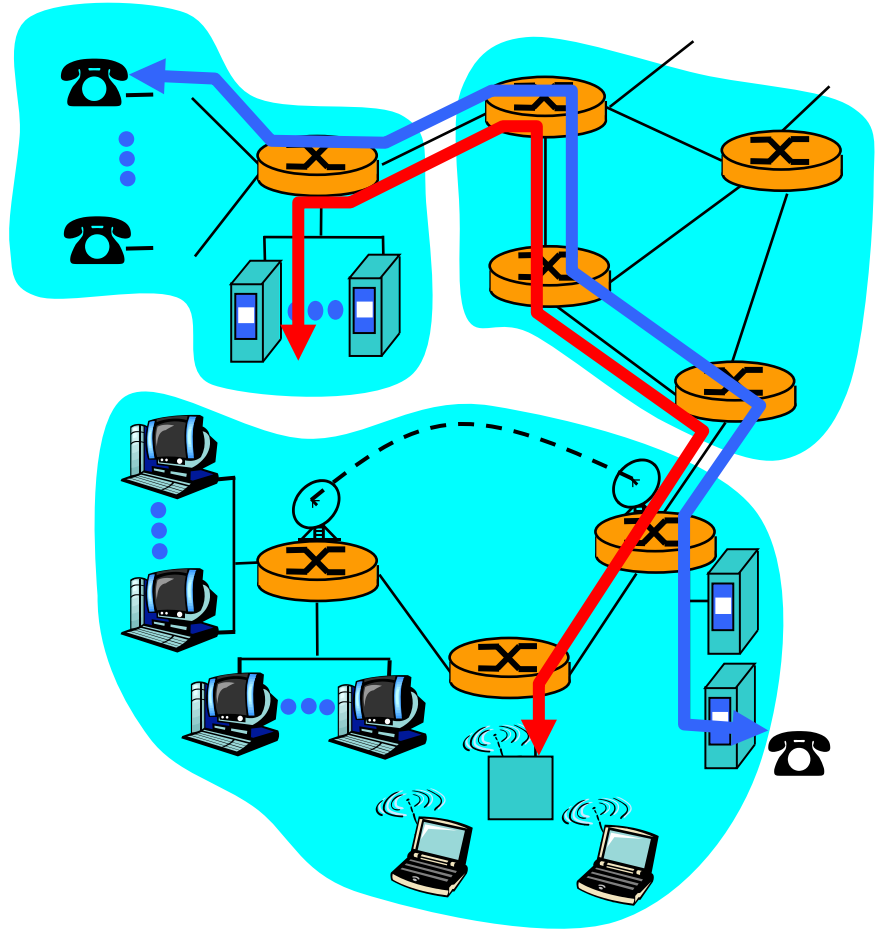


Circuit Switching

End-end circuit reserved
for a “call”

Resources needed for
communication along
a path are reserved

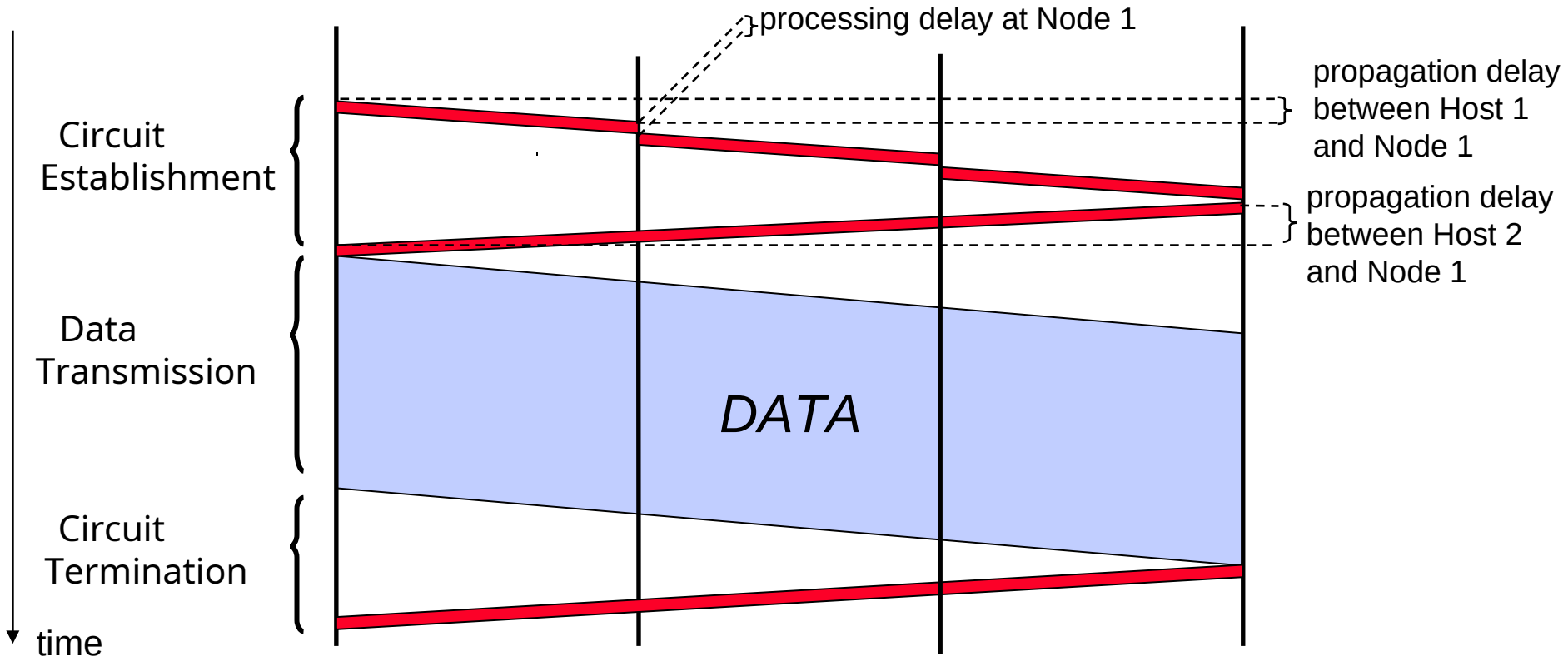
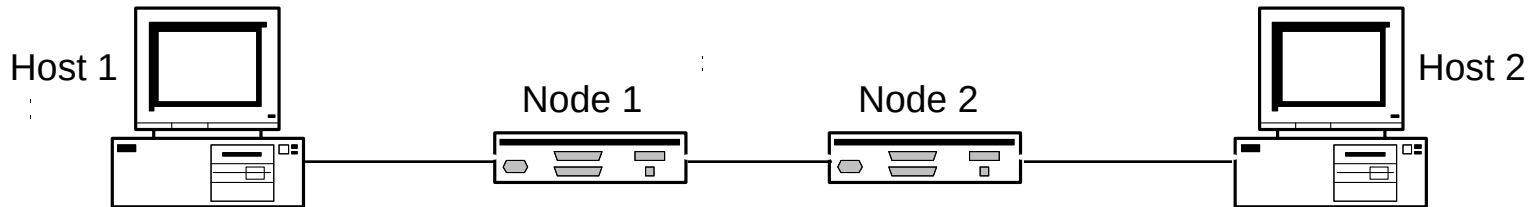
- transmission capacity,
buffer etc.



Circuit Switching

- Three phases
 1. circuit establishment
 2. information transfer
 3. circuit termination
- If circuit not available: “Busy signal”
- Examples
 - Telephone networks

Timing in Circuit Switching



Resource Division

Network resources (e.g., bandwidth) **divided into “pieces”**

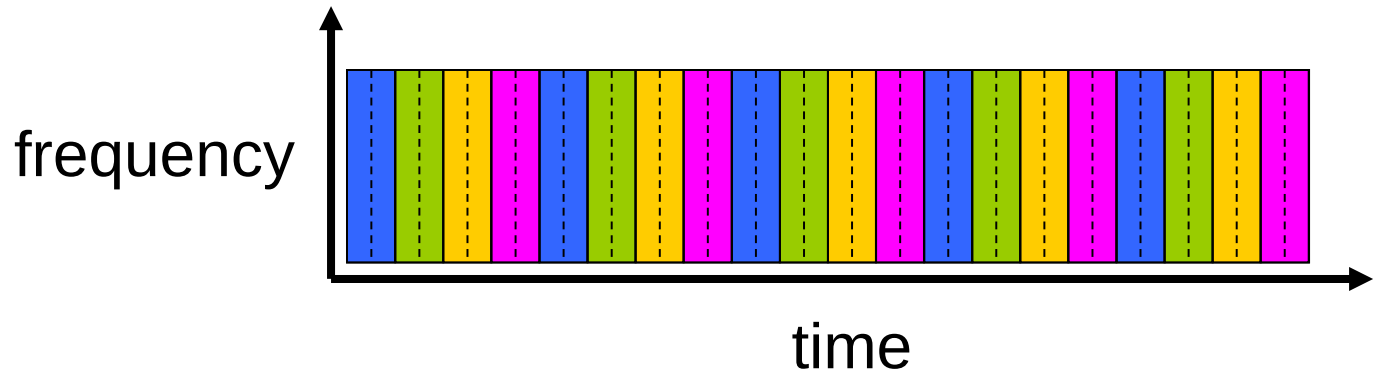
- pieces allocated to calls
- resource piece *idle* if not used by owning call (*no sharing*)
- dividing link bandwidth into “pieces”
 - time division
 - frequency division

Circuit Switching: TDM and FDM

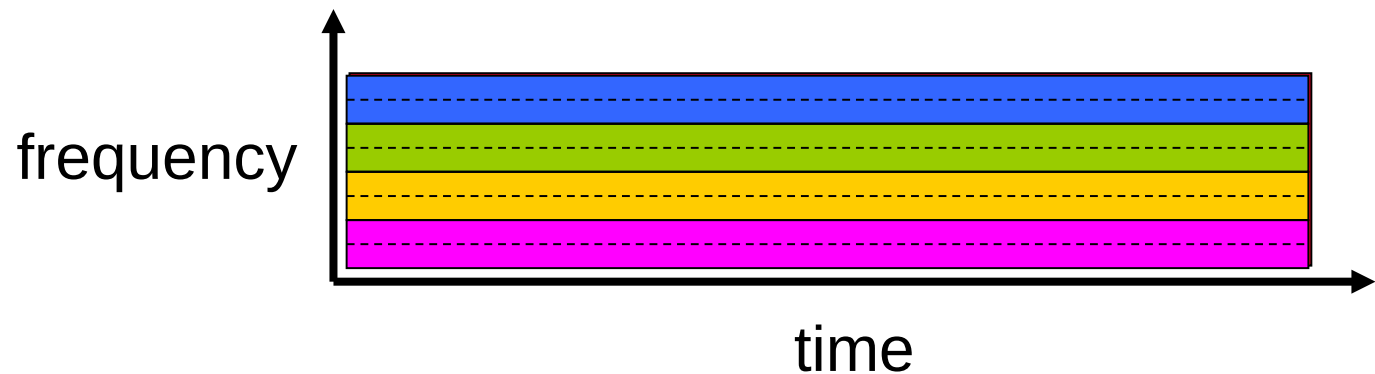
TDMA

Example:

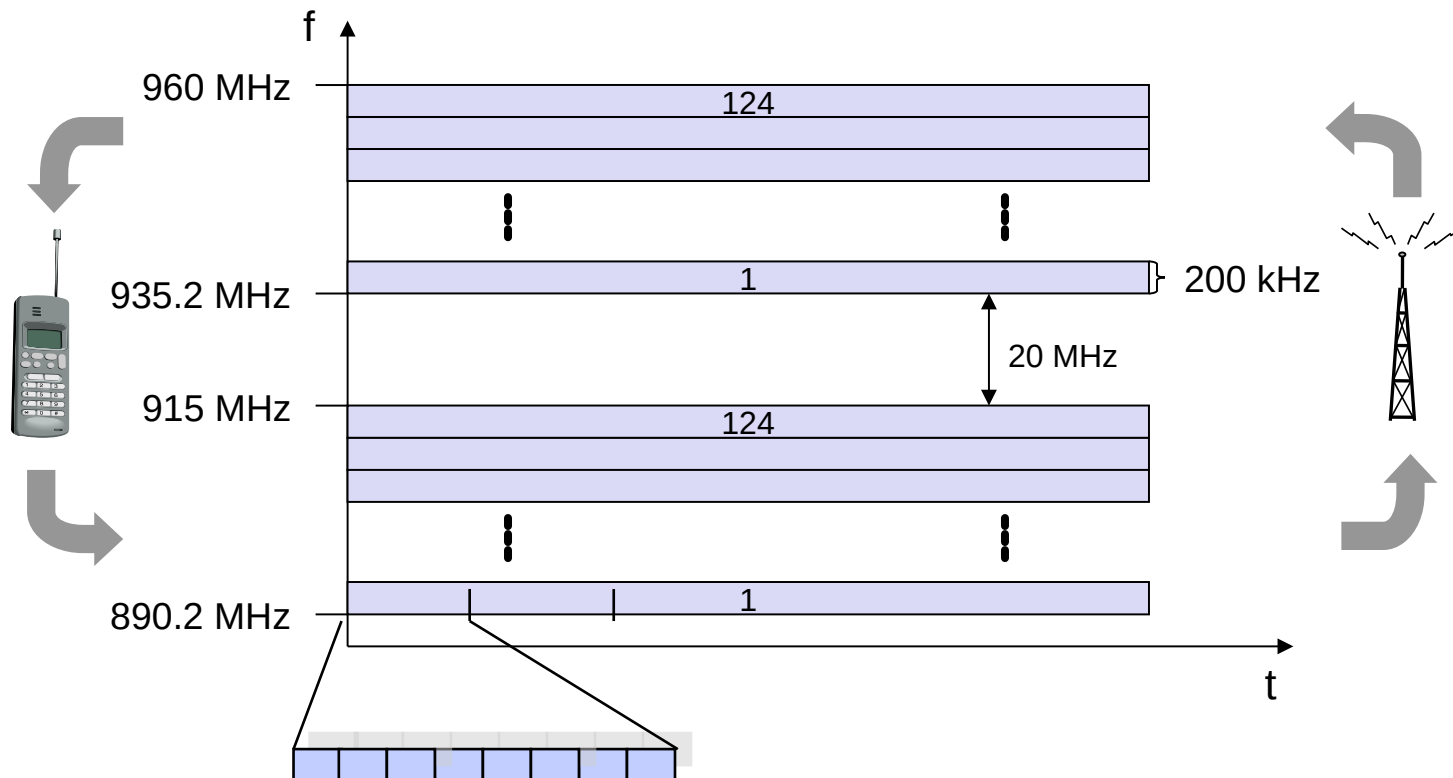
4 users



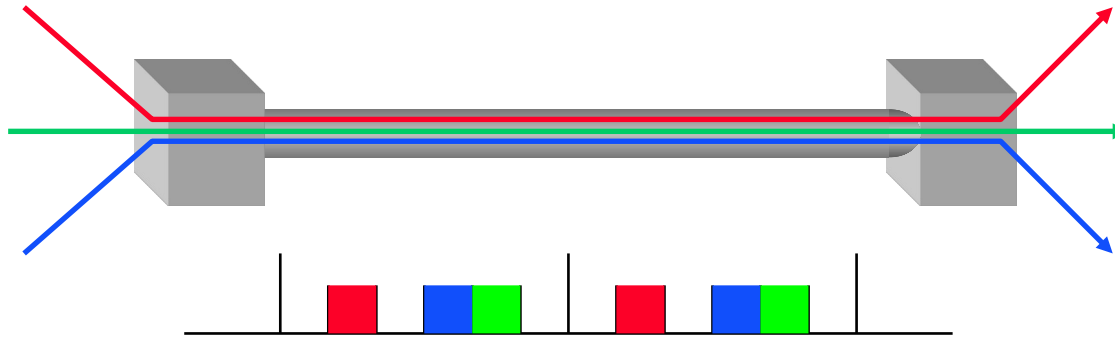
FDMA



FDM/TDM - example GSM 900



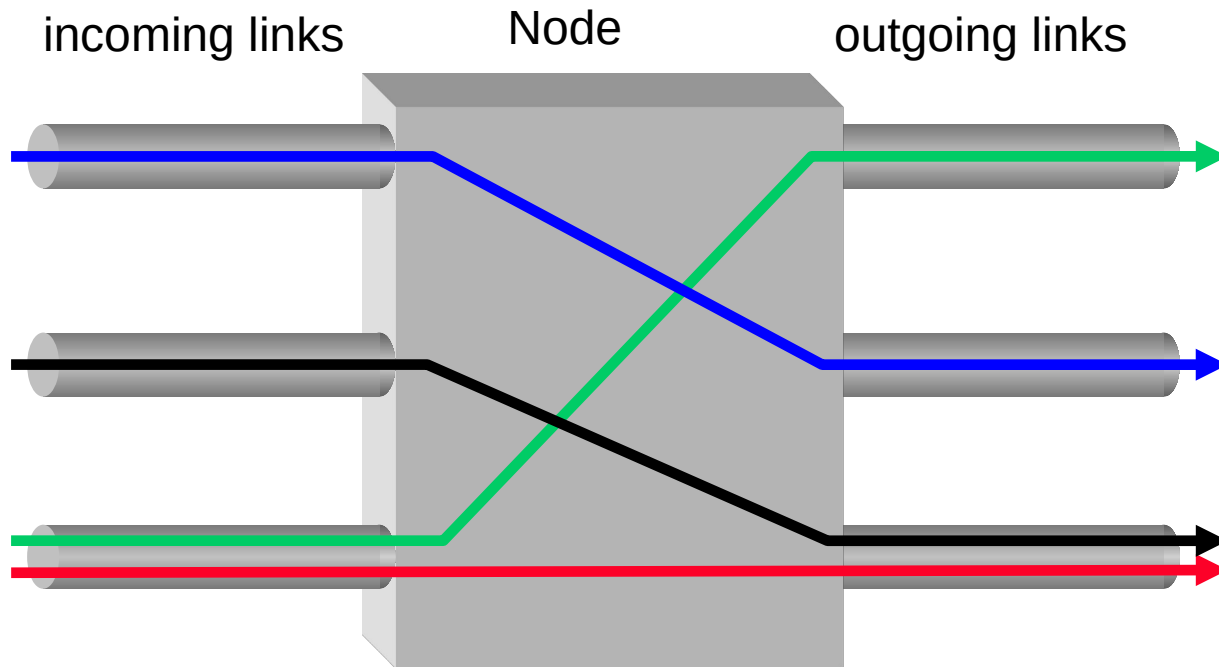
Circuit Switching: Multiplexing/Demultiplexing



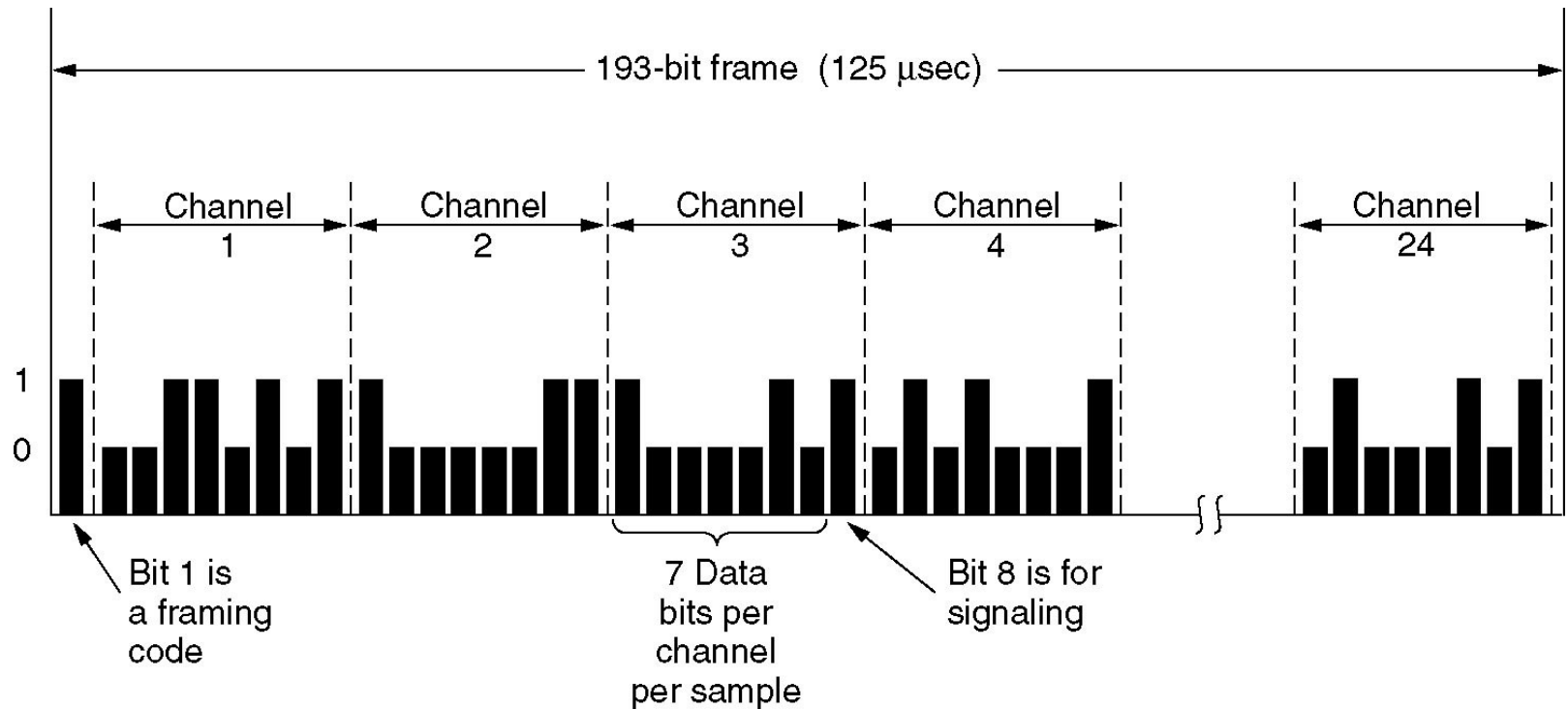
- Time divided in frames and frames divided in slots
- Relative slot position inside a frame **determines** which circuit the data uses
- Needs synchronization between sender and receiver

Circuit Switching

- A node (switch) in a circuit switching network



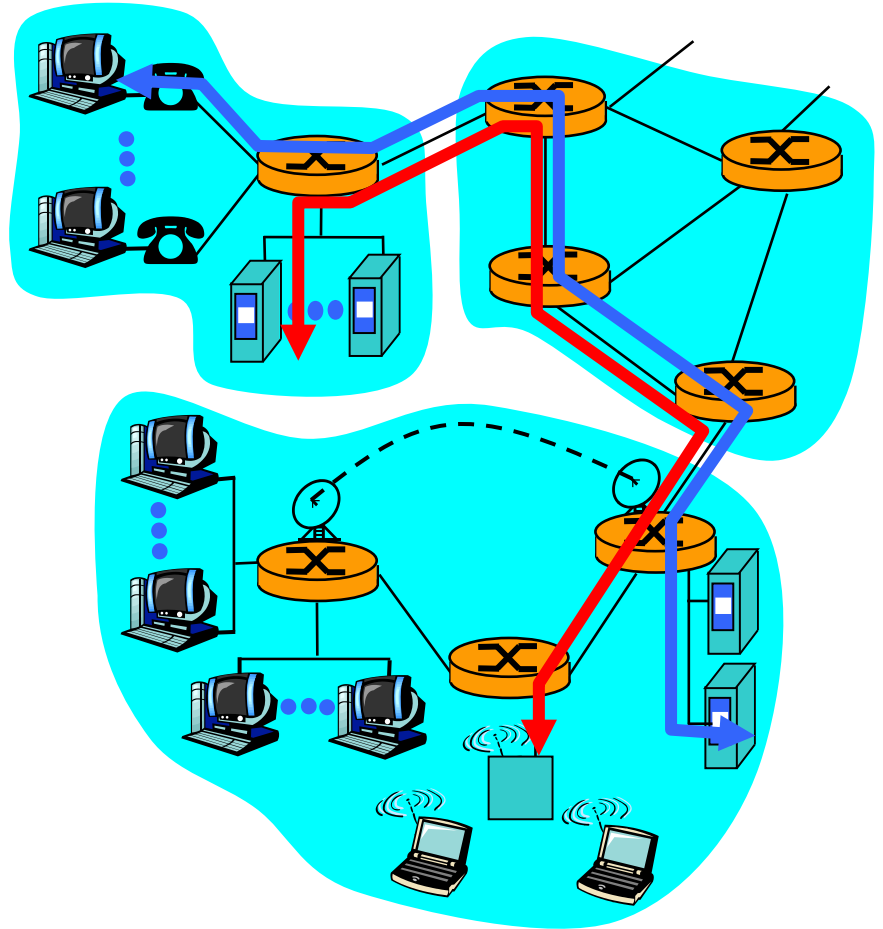
Time Division Multiplexing



Circuit Switching - Summary

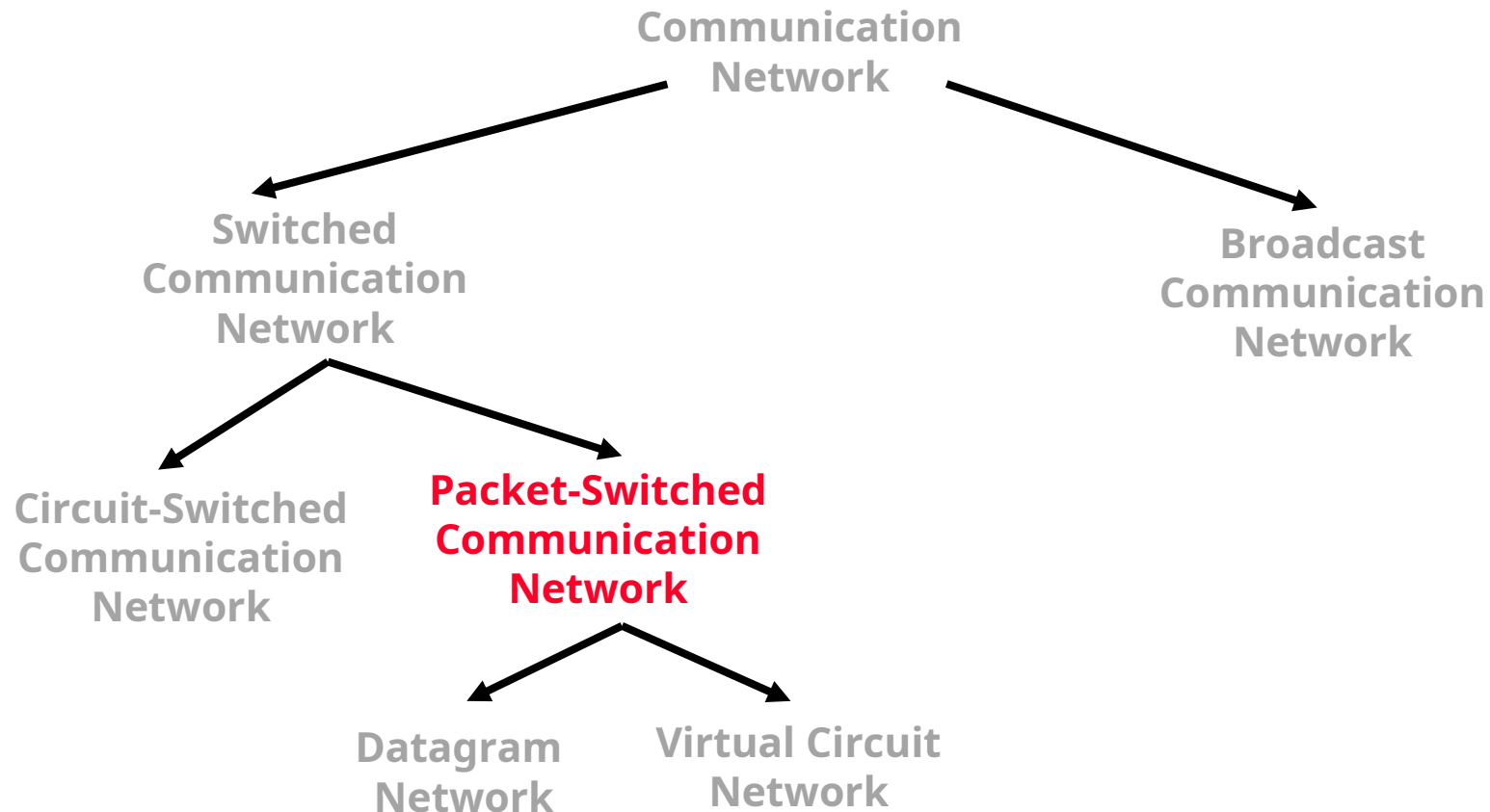
End-end resources
reserved for a “call”

- call setup required
- link bandwidth, switch capacity
- dedicated resources: no sharing
- circuit: guarantee performance



A Taxonomy of Communication Networks

- Communication networks can be classified based on the way in which the nodes exchange information:



Packet Switching

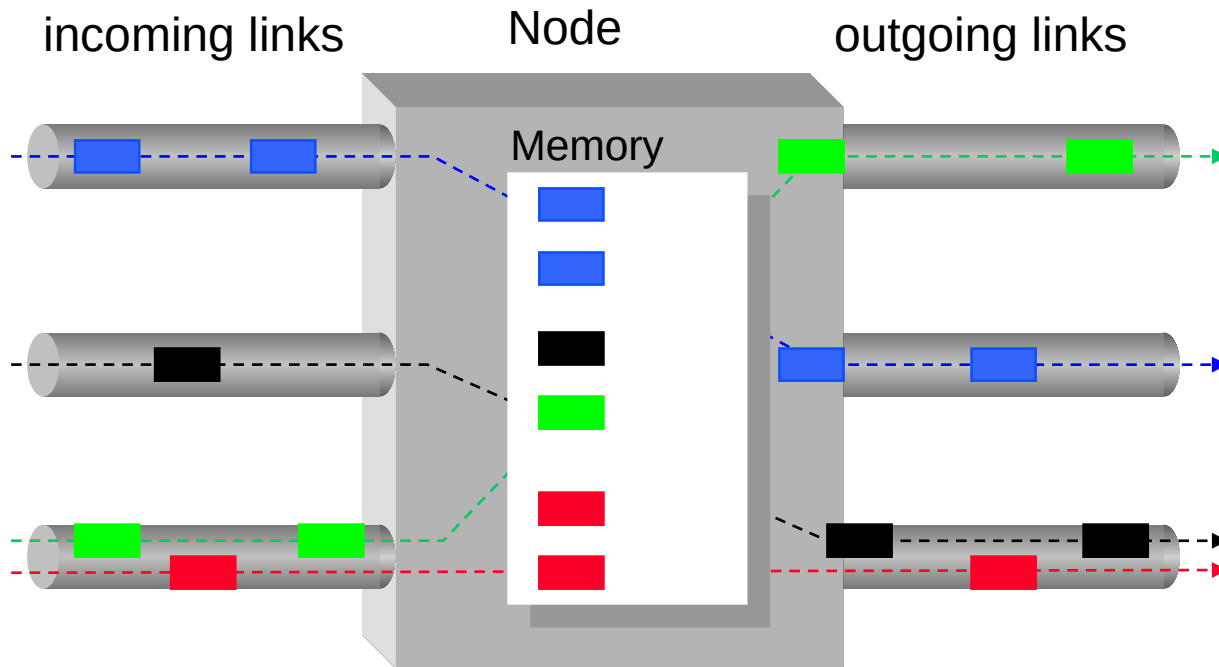
- Data are sent as formatted bit-sequences, so-called **packets**
- Packets have the following structure:



- **Header** and **Trailer** carry control information (e.g., destination address, checksum)
- **Routing:** Each packet is passed through the network from node to node along some path
- **Store-and-Forward:** At each node the entire packet is received, stored briefly, and then forwarded to the next node
- No resource is reserved for packets

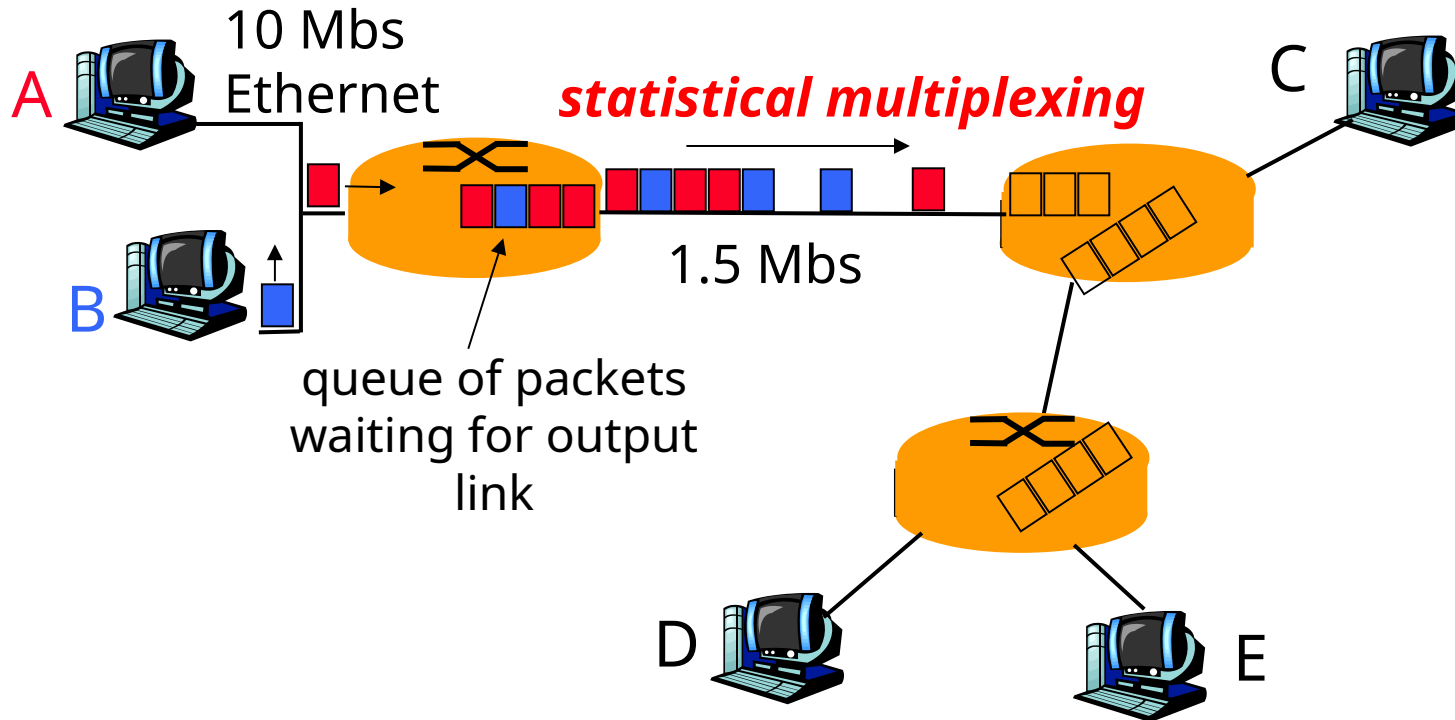
Packet Switching

- A node in a packet switching network



Store and forward

Packet Switching: Statistical Multiplexing



Sequence of A & B packets does not have a fixed pattern ➡
statistical multiplexing.

Packet Switching - Summary

- resources used *as needed*

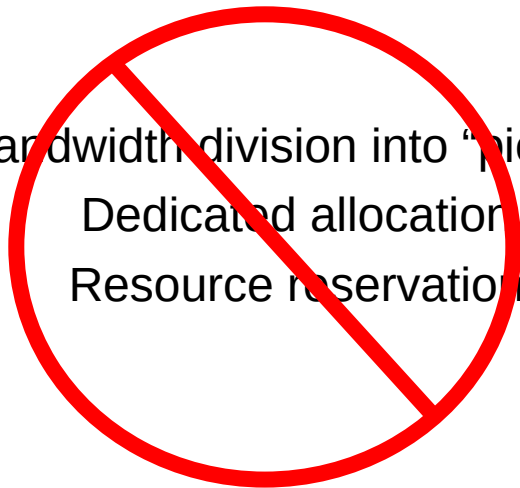
resource contention:

- store and forward**: packets move one hop at a time
 - transmit over link
 - wait turn at next link
- aggregate resource demand can exceed amount available
- congestion: packets queue, wait for link use
 - packet loss

Bandwidth division into “pieces”

Dedicated allocation

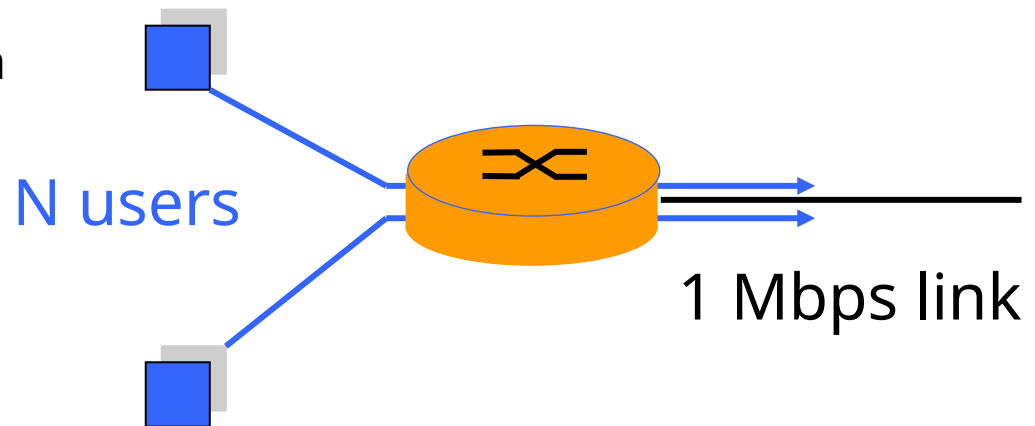
Resource reservation



Packet switching versus circuit switching

Packet switching allows more users to use network!

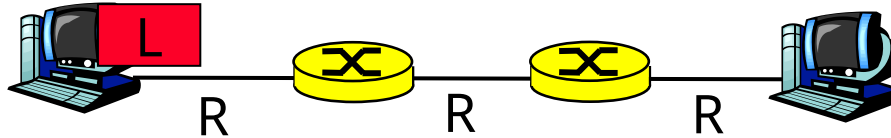
- 1 Mbits/sec link
- each user:
 - 100 kbps when “active”
 - active 10% of time or with probability 0.1
- circuit-switching:
 - 10 users
- packet switching:
 - with 35 users
 - probability > 10 active less than .0004



Packet switching versus circuit switching

- Great for bursty data
 - resource sharing
 - no call setup
- Congestion may happen in packet switching:
 - packet delay and loss
 - protocols needed for reliable data transfer, congestion control

Packet-switching: store-and-forward

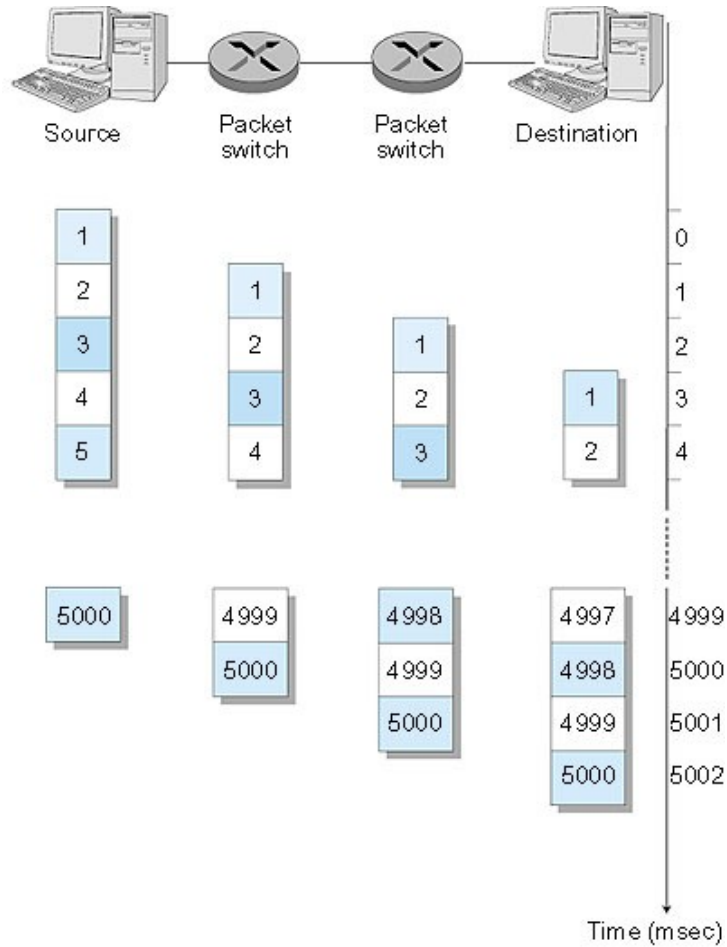


- Takes L/R seconds to transmit (push out) packet of L bits on to link of R bps
- Entire packet must arrive at router before it can be transmitted on next link: *store and forward*
- Transmission delay = $3L/R$

Example:

- $L = 7.5$ Mbits
- $R = 1.5$ Mbps
- Transmission delay = 15 sec

Packet Switching: Message Segmenting



Now break up the message
(7.5Mbits) into 5000 packets

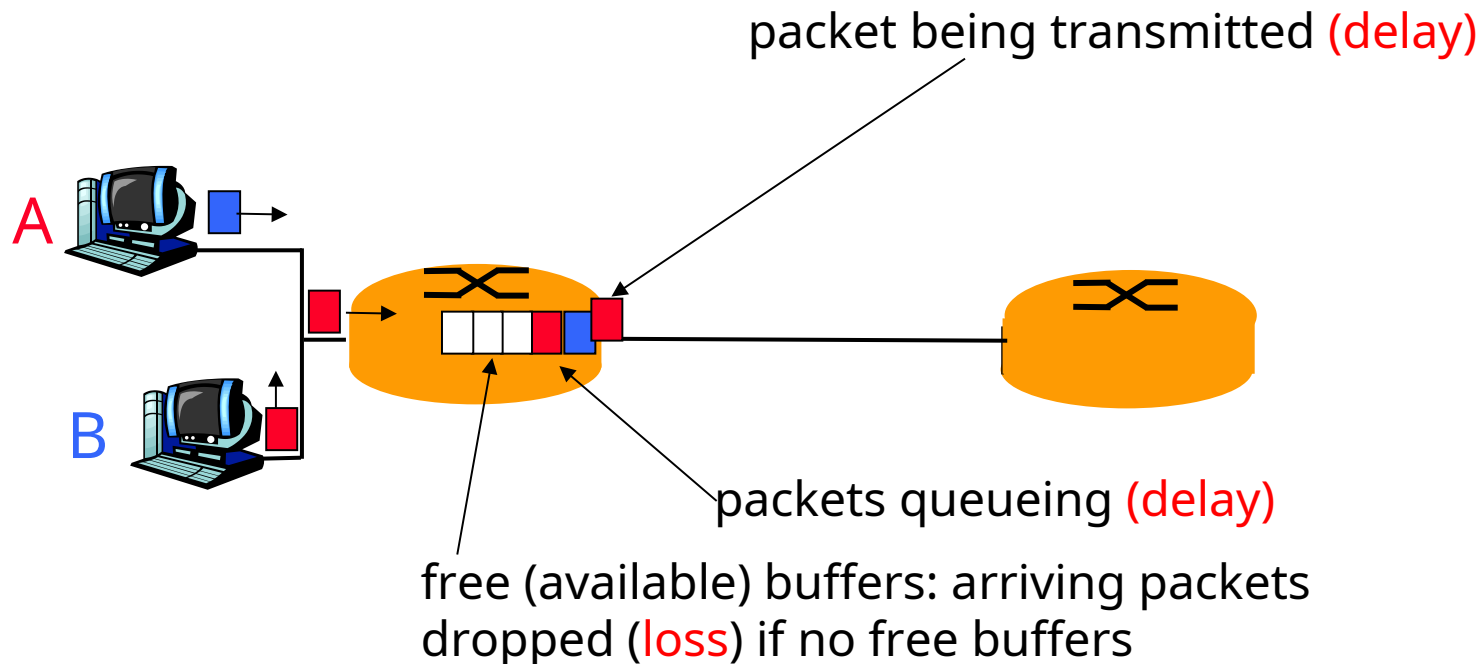
- Each packet 1,500 bits
- 1 msec to transmit packet on one link
- *pipelining*: each link works in parallel
- Transmission delay reduced from 15 sec to 5.002 sec

Delay & loss in packet-switched networks

How do loss and delay occur?

Packets *queue* in router buffers

- packet arrival rate to link exceeds output link capacity
- packets queue, wait for turn

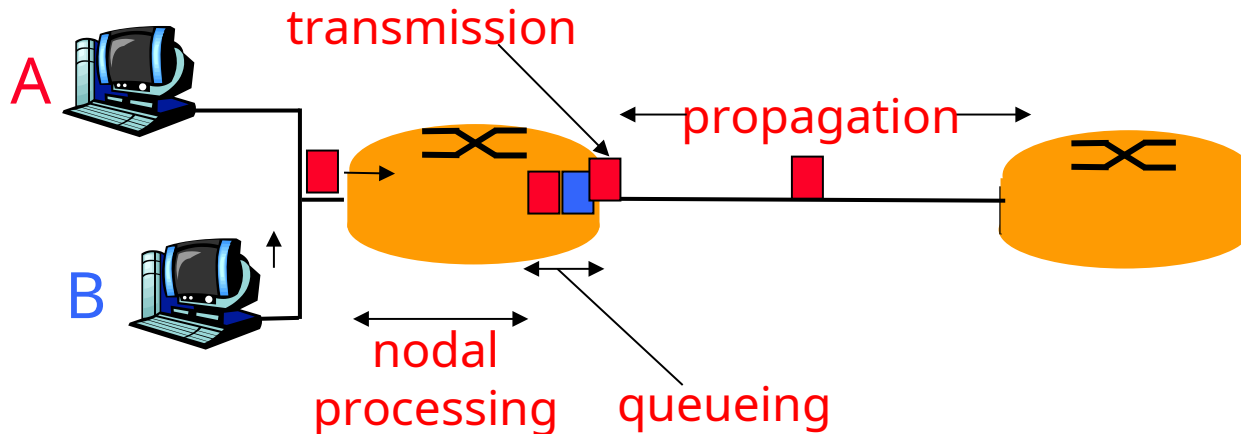


Packet loss

- Queue (aka buffer) preceding link has finite capacity
- When packet arrives to full queue, packet is dropped (aka lost) – buffer overflow
- Lost packet may be retransmitted by previous node, by source end system, or not retransmitted at all

Four sources of packet delay

- 1. nodal processing:
 - check bit errors
 - determine output link
- 2. queueing
 - time waiting at output link for transmission
 - depends on congestion level of router



Delay in packet-switched networks

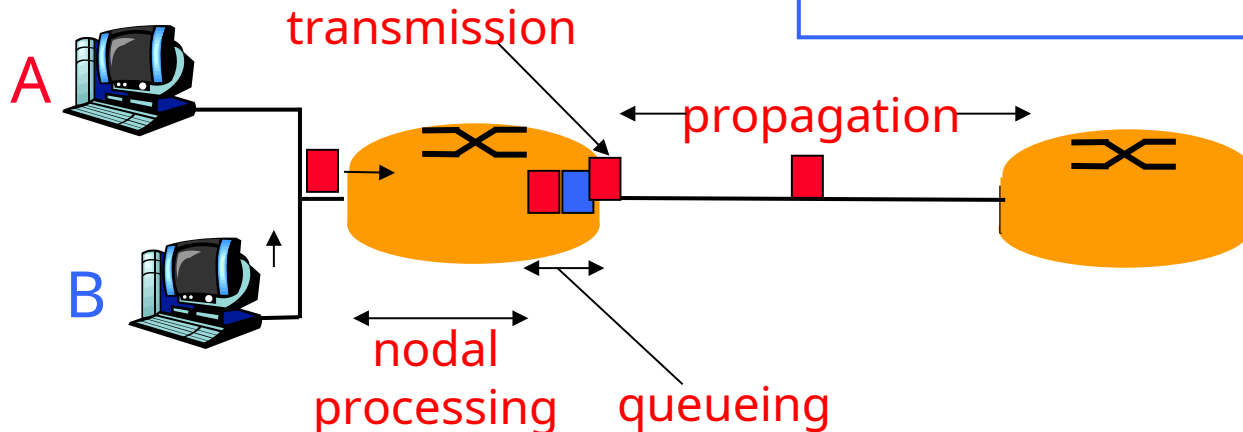
3. Transmission delay:

- R = link bandwidth (bps)
- L = packet length (bits)
- time to send bits into link = L/R

4. Propagation delay:

- d = length of physical link
- s = propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- propagation delay = d/s

Note: s and R are very different quantities!

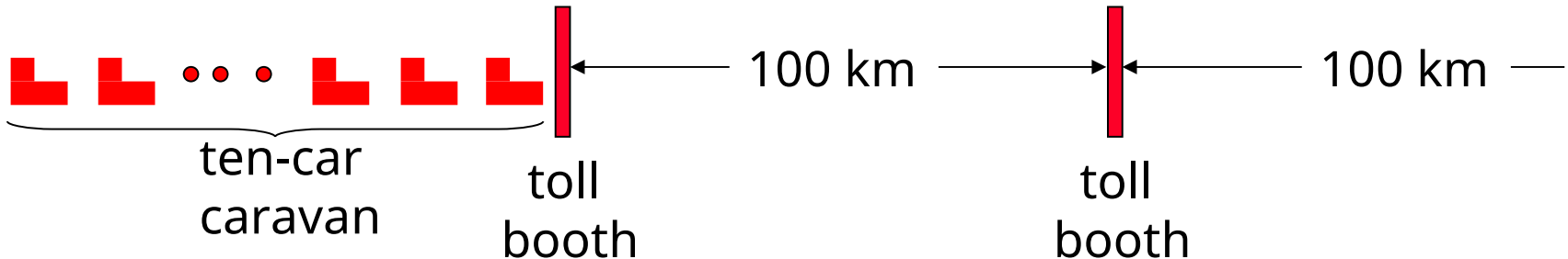


Nodal delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

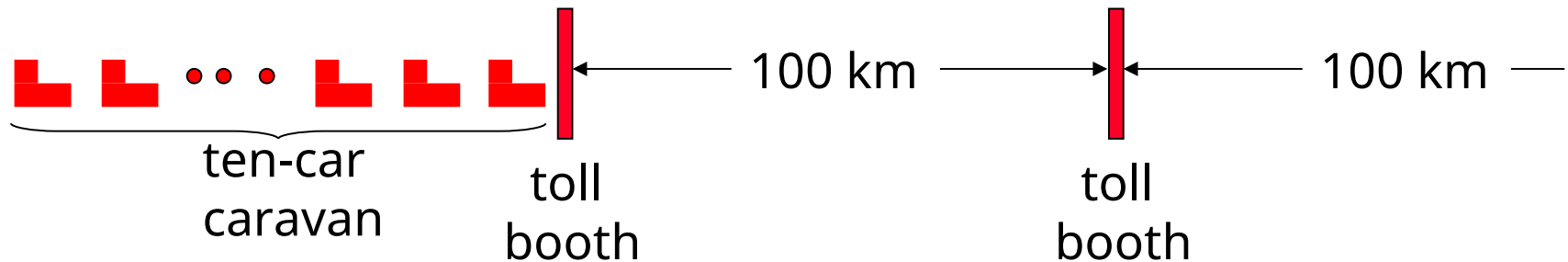
- d_{proc} = processing delay
 - typically a few microsecs or less
- d_{queue} = queuing delay
 - depends on congestion condition
- d_{trans} = transmission delay
 - $= L/R$, significant for low-speed links
- d_{prop} = propagation delay
 - a few microsecs to hundreds of msecs

Caravan analogy



- Cars “propagate” at 100 km/hr
- Toll booth takes 12 sec to service a car (transmission time)
- car~bit; caravan ~ packet
- Q: How long until caravan is lined up before 2nd toll booth?
- Time to “push” entire caravan through toll booth onto highway = $12 \times 10 = 120$ sec
- Time for last car to propagate from 1st to 2nd toll booth: $100\text{km}/(100\text{km/hr}) = 1$ hr
- A: 62 minutes

Caravan analogy (more)

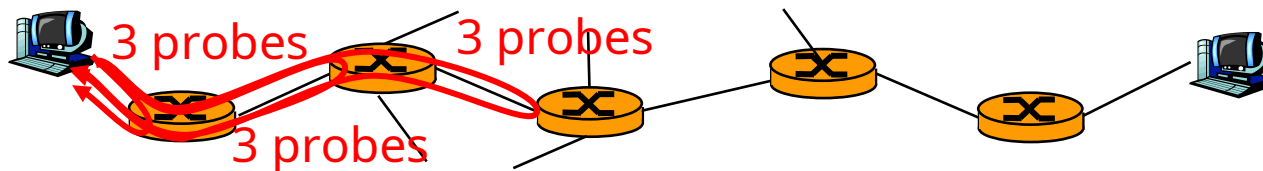


- Cars now “propagate” at 1000 km/hr
- Toll booth now takes 1 min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at 1st booth?**

- **Yes!** After 7 min, 1st car at 2nd booth and 3 cars still at 1st booth.
- **1st bit of packet can arrive at 2nd router before packet is fully transmitted at 1st router!**

Internet delays and routes

- What do Internet delay & loss look like?
- **Traceroute program:** provides delay measurement from source to router along end-end Internet path towards destination. For each intermediate router i :
 - sends three packets that will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply



Delay and routes

- ping hostname
- traceroute gamma
- traceroute www.cs.wright.edu
- traceroute www.wright.edu
- traceroute www.osu.edu
- traceroute www.bbcnews.co.uk

Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

Three delay measurements from
gaia.cs.umass.edu to cs-
gw.cs.umass.edu



```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
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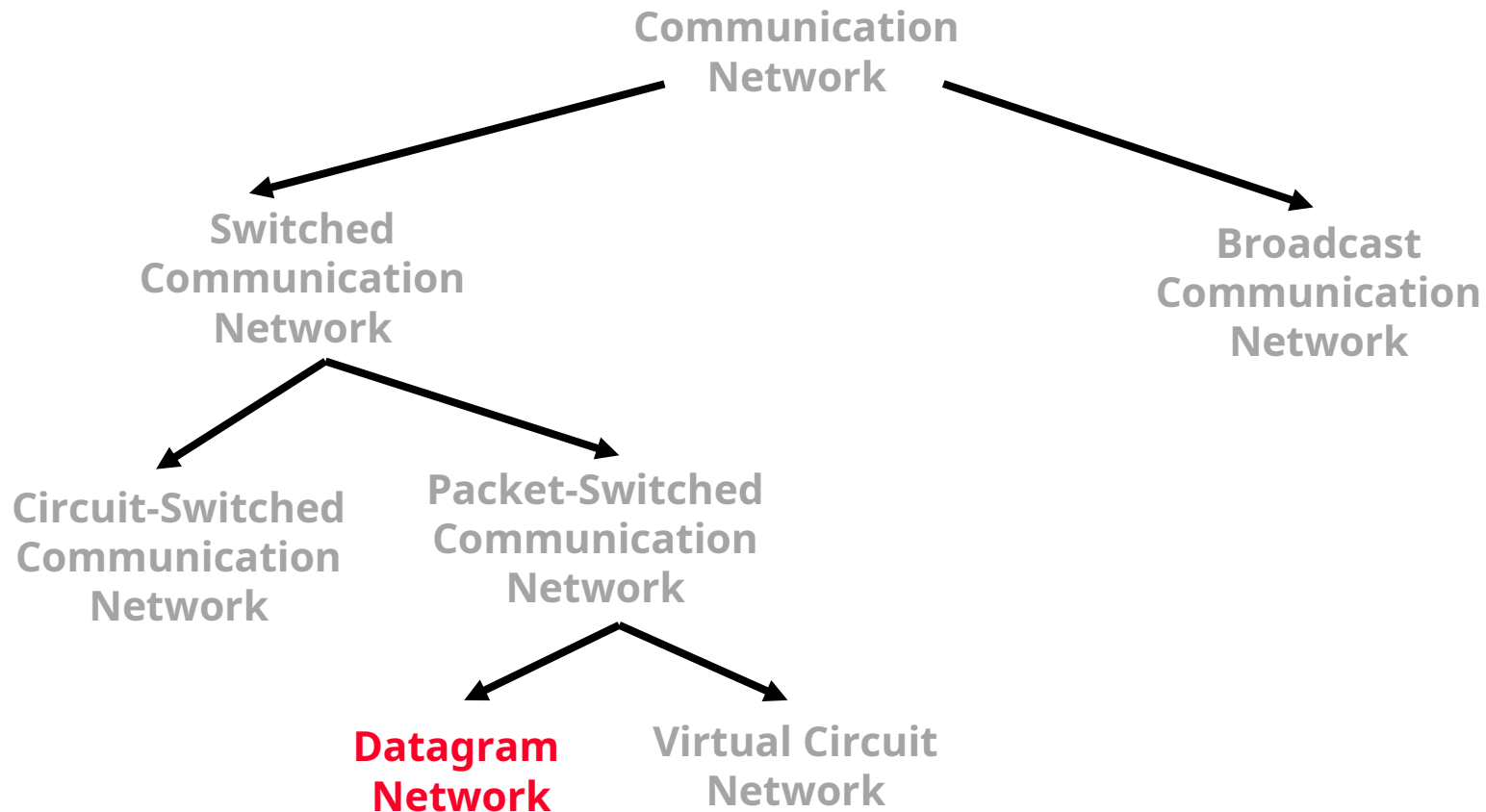
trans-oceanic link

* means no response (probe lost, router not replying)



A Taxonomy of Communication Networks

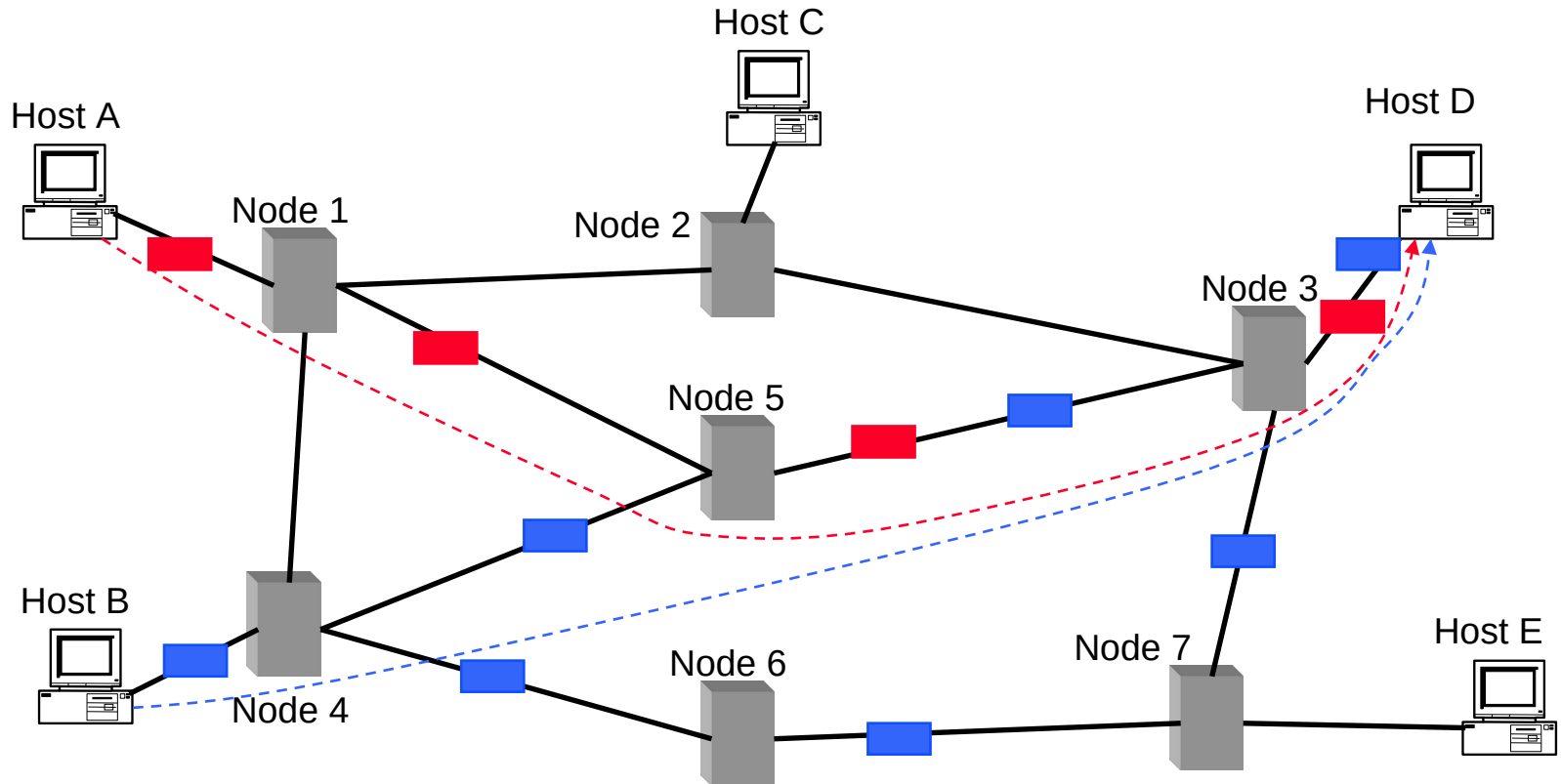
- Communication networks can be classified based on the way in which the nodes exchange information:



Datagram Packet Switching

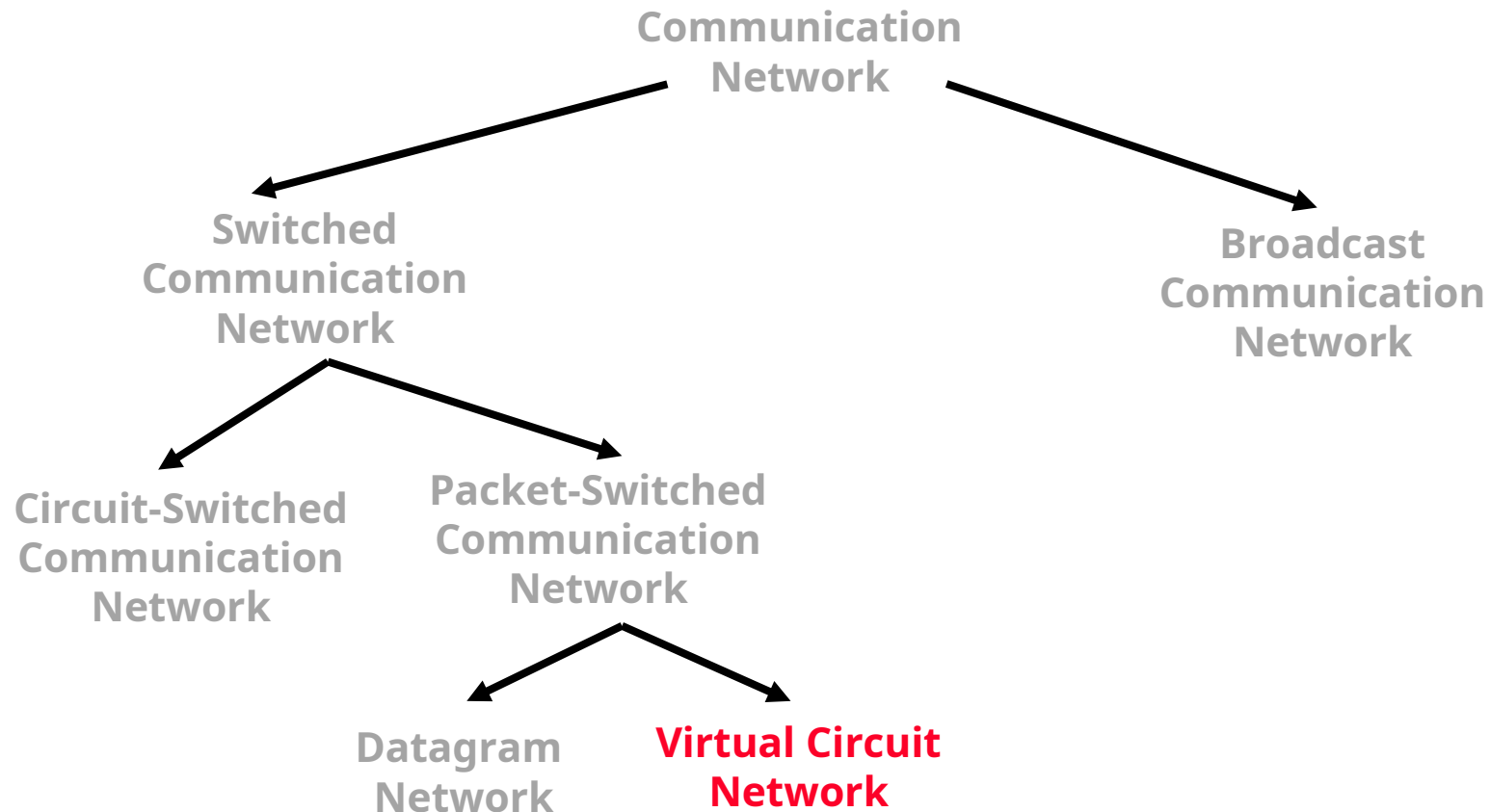
- Each packet is independently forwarded
 - based on destination address
 - each packet header contains destination address
- No resources are pre-allocated (reserved) in advance
- Example: IP networks

Datagram Packet Switching

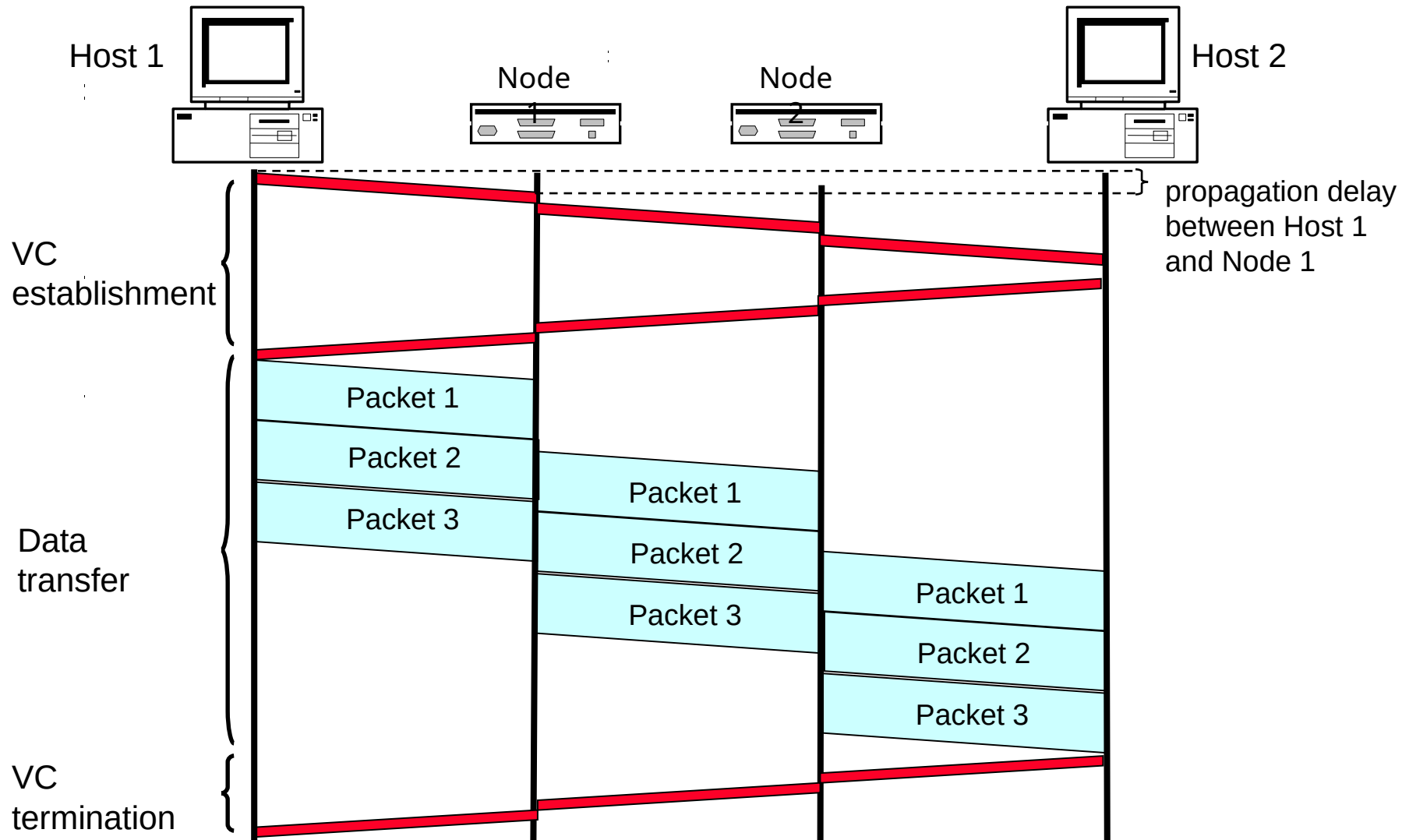


A Taxonomy of Communication Networks

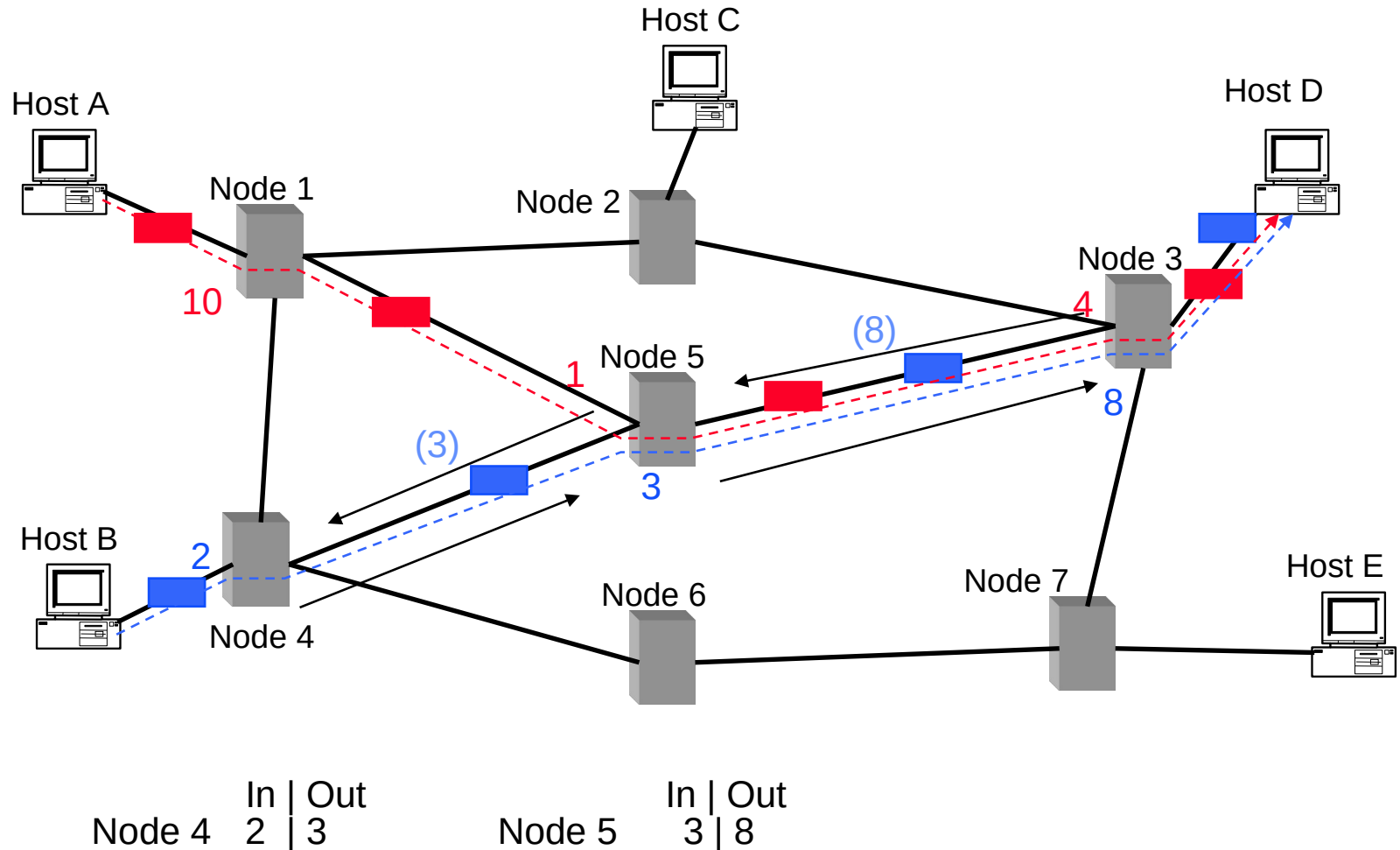
- Communication networks can be classified based on the way in which the nodes exchange information:



Timing of VC Packet Switching



Virtual Circuit Packet Switching



Virtual-Circuit Packet Switching

- Conceptually a hybrid of circuit switching and packet switching
 - all packets from one packet stream are sent along a pre-established path (= virtual circuit)
 - **no resources are reserved – still packet switching**
- Guarantees packets use the same path
- **However:** Packets from different virtual circuits may be interleaved – due to *statistical multiplexing*
- Example: ATM networks (see a short introduction to ATM networks posted in WebCT)

Virtual-Circuit Packet Switching

- Communication with virtual circuits takes place in three phases
 1. VC establishment
 2. data transfer
 3. VC disconnect
- VC identifier: VCI
- Note: packet headers don't need to contain the full destination address of the packet

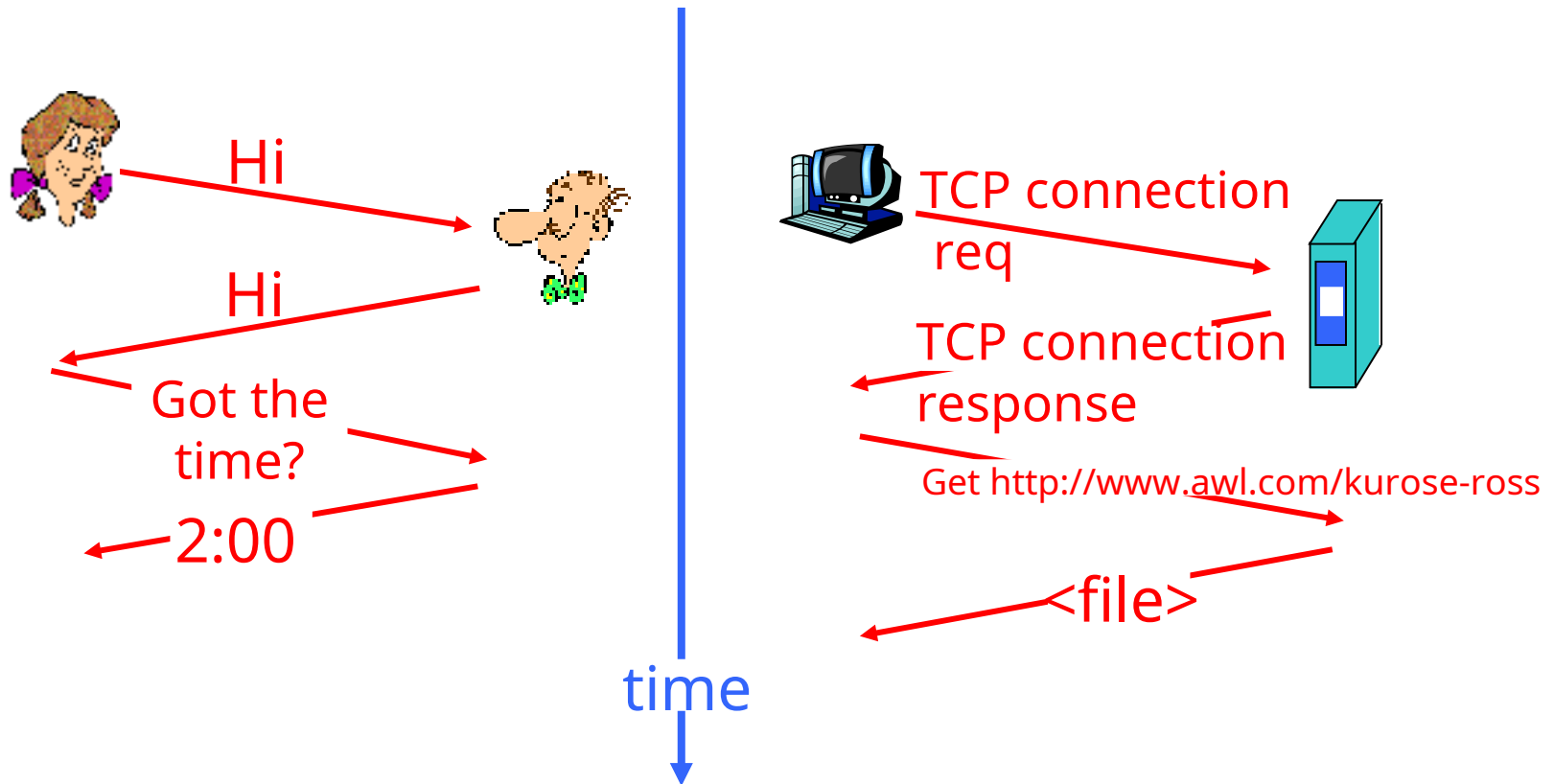
Datagram vs Virtual Circuit

- **datagram network:**
 - *destination address* in packet determines next hop
 - routes may change during session
 - analogy: driving, asking directions
- **virtual circuit network:**
 - each packet carries tag (virtual circuit ID), tag determines next hop
 - fixed path determined at *call setup time*, remains fixed throughout call
 - *routers maintain per-call state*

Protocol, protocol layering and layers

What's a protocol?

a human protocol and a computer network protocol:



What's a protocol?

human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific messages sent

... specific actions taken
when messages received,
or other events

network protocols:

- machines rather than humans
- all communication activity in Internet governed by protocols

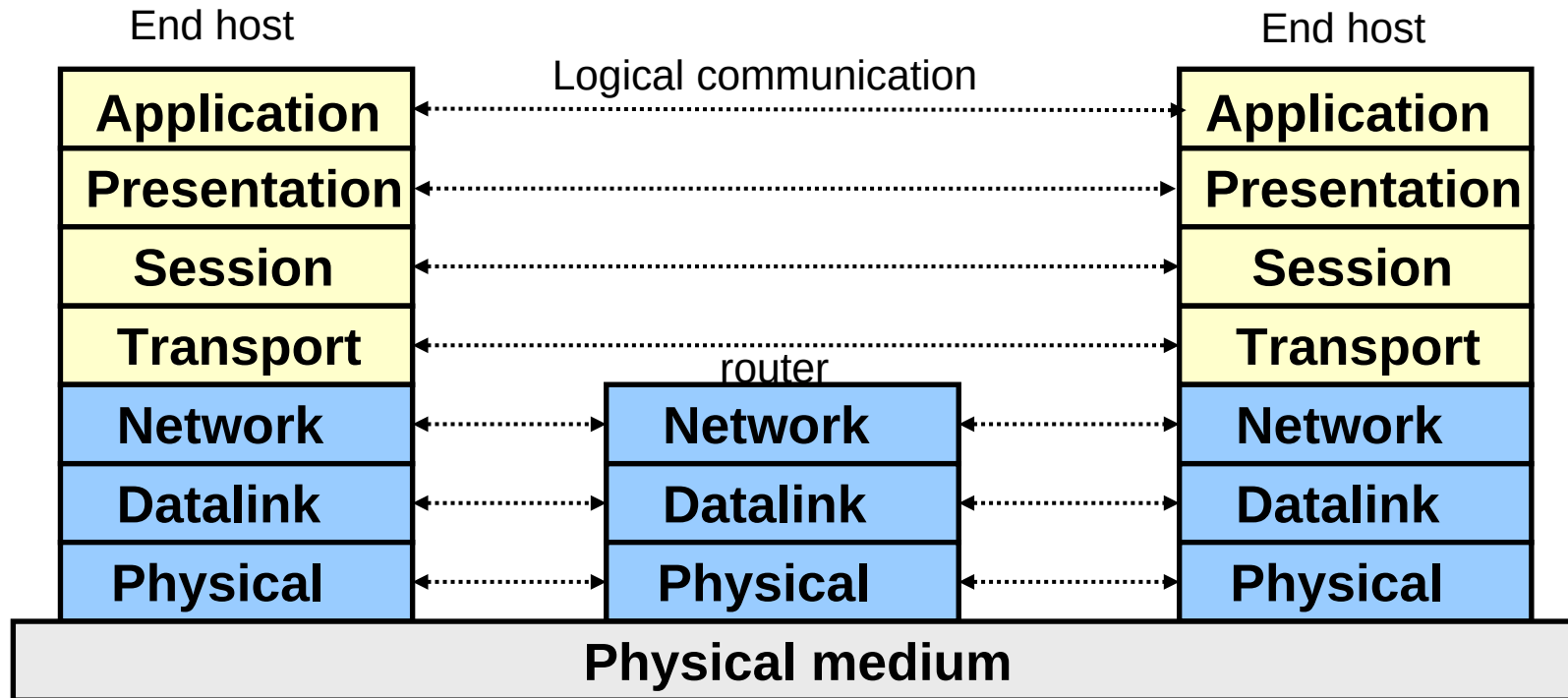
protocols define format, order of messages sent and received among network entities, and timing of actions and actions taken on message transmission/receipt, or other events

OSI Reference Model

- Seven layers

- Lower three layers are *peer-to-peer*
- Next four layers are *end-to-end*

Important

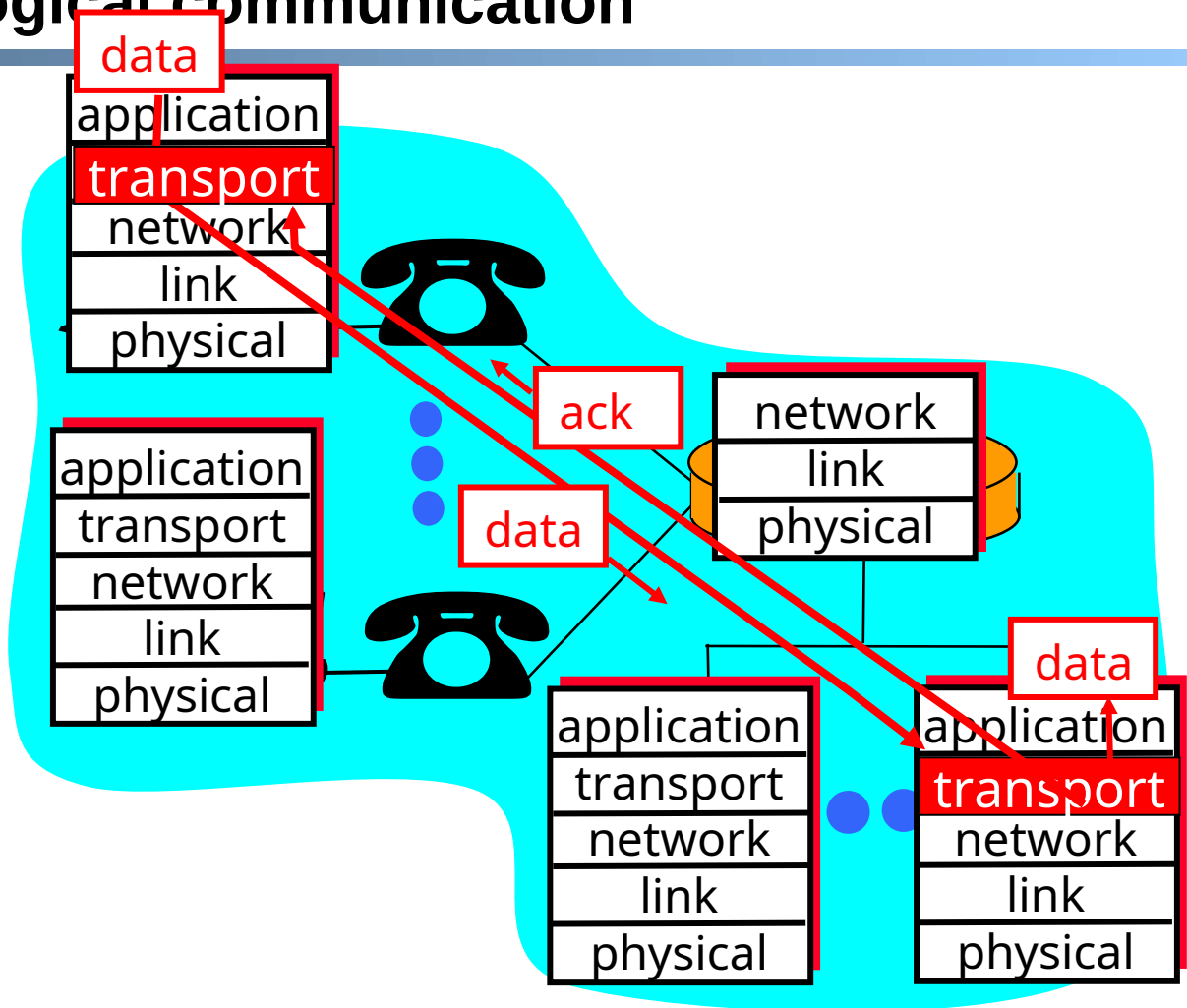


ISO: international standard organization

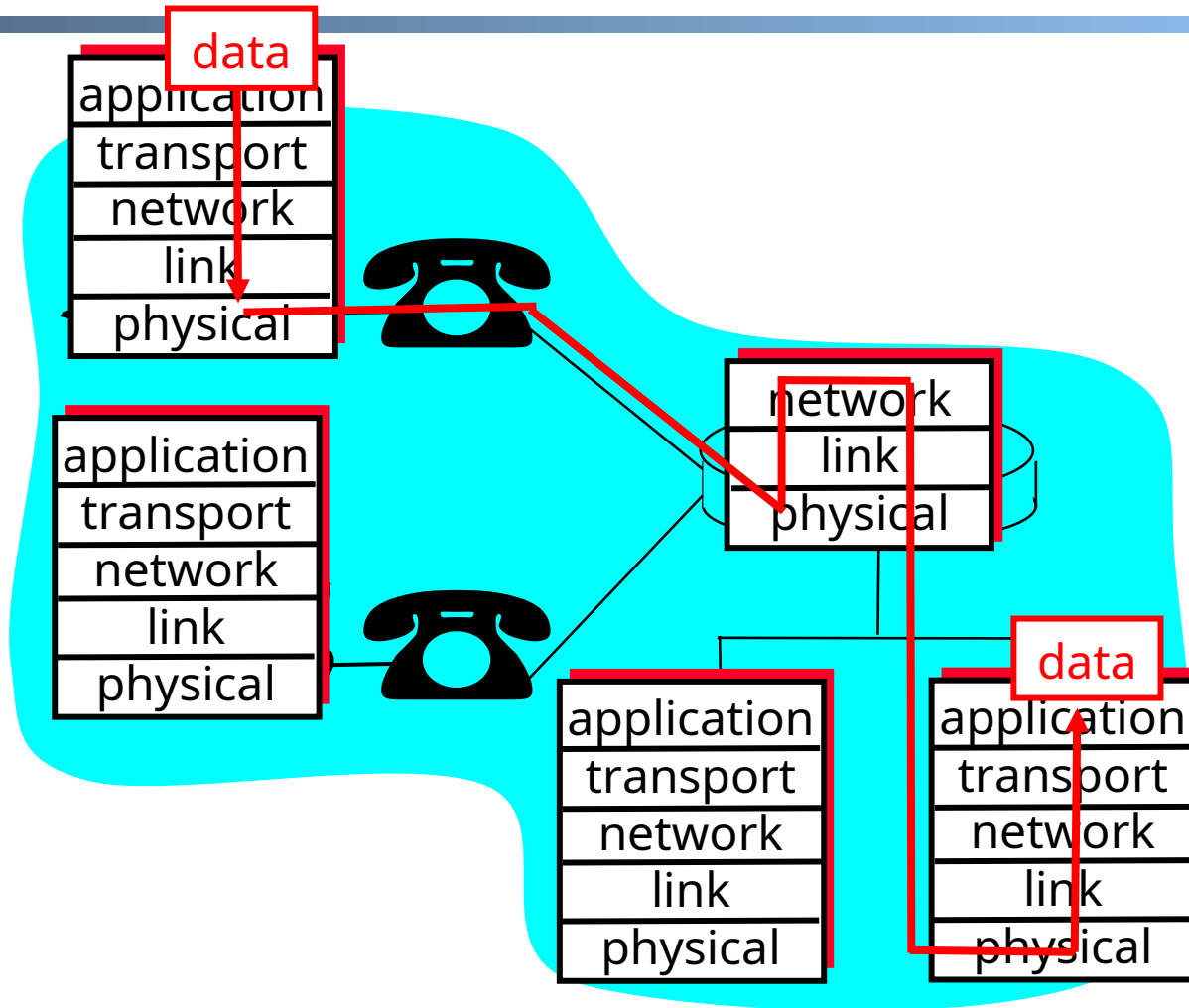
Logical communication

E.g.: transport

- take data from app
- add addressing, reliability check info to form “datagram”
- send datagram to peer
- wait for peer to ack receipt
- analogy: post office

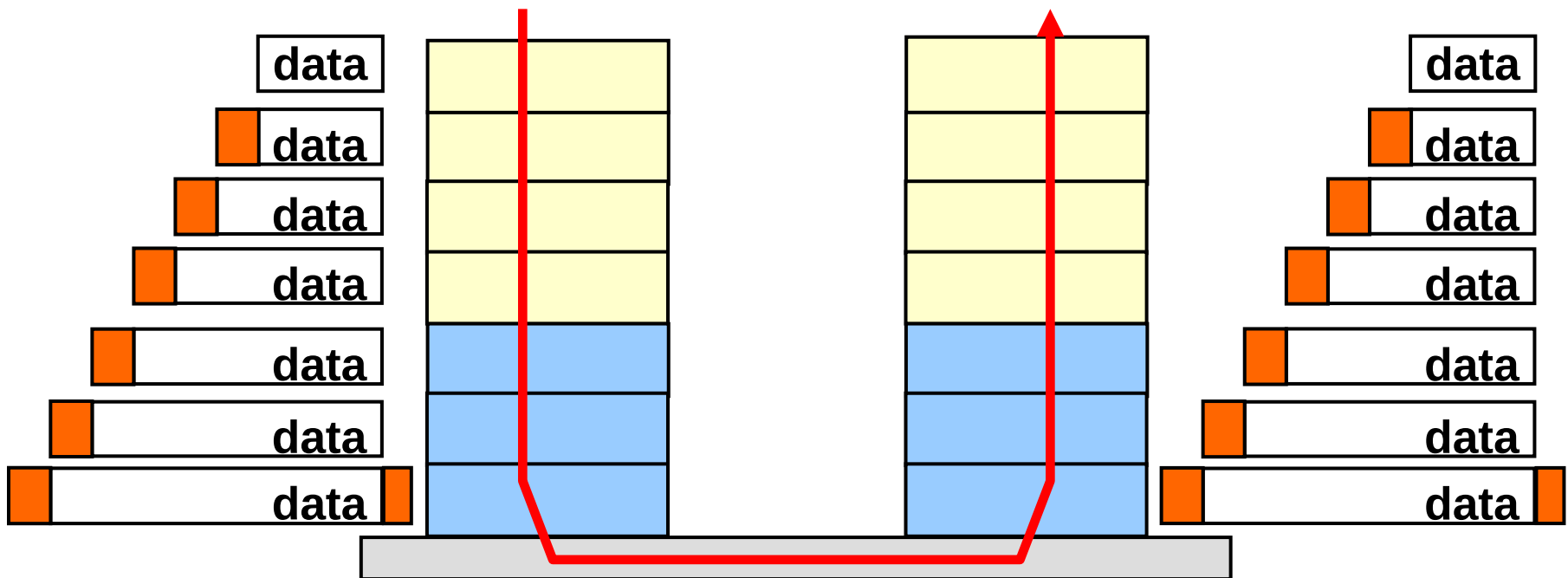


Physical communication



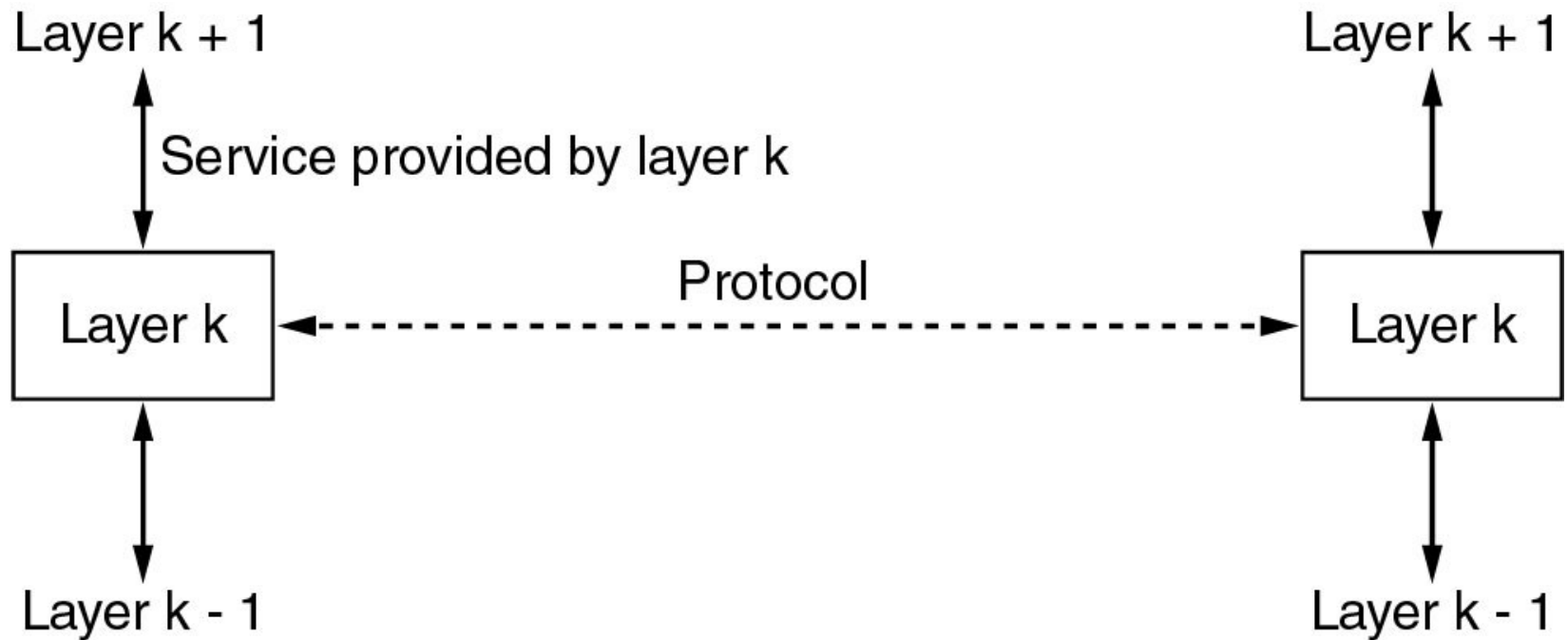
Data Transmission

- A layer can use **only** the service provided by the layer immediate below it
- Each layer may add a new header to data packet

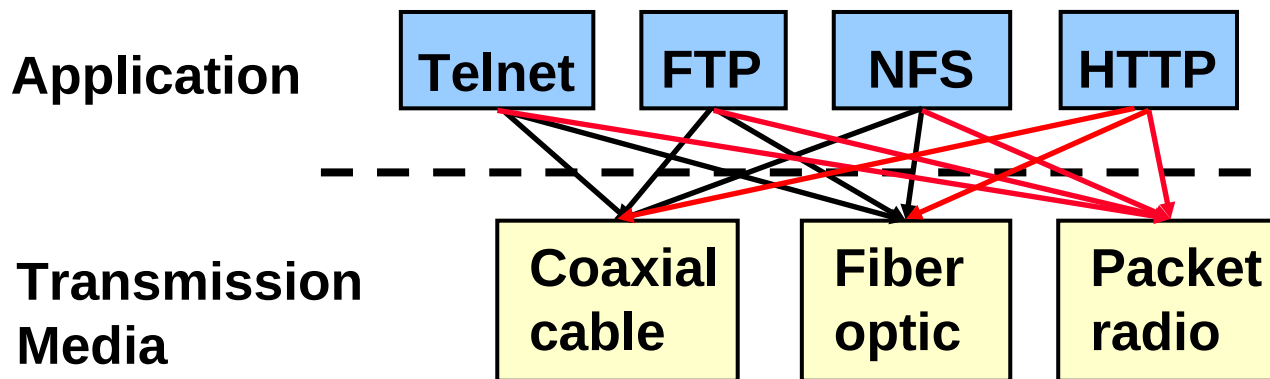


Services to Protocols Relationship

- The relationship between a service and a protocol.



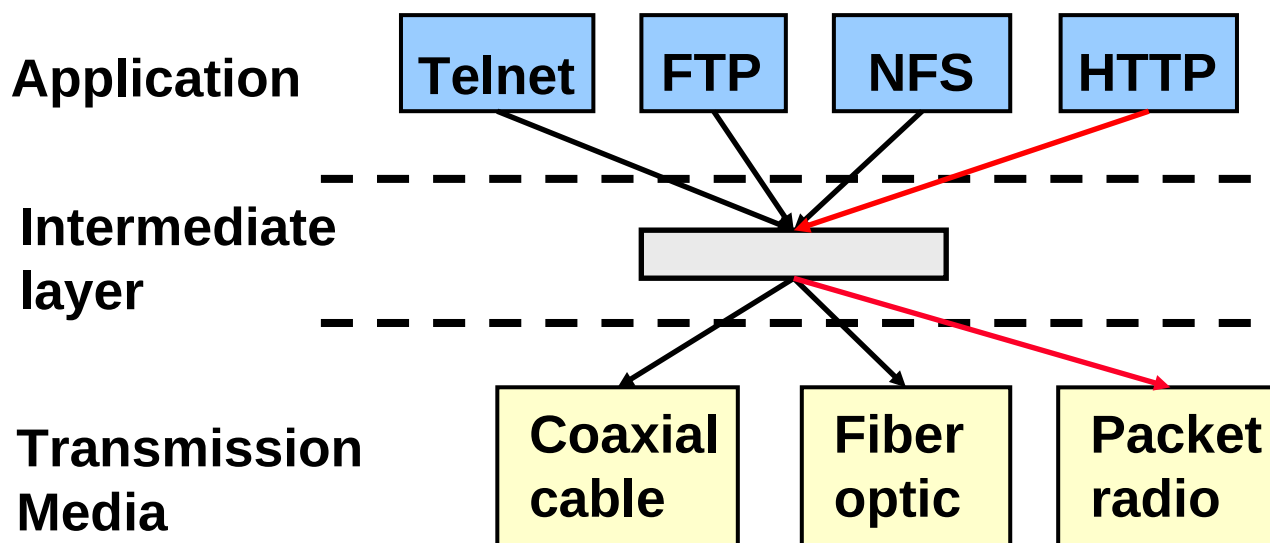
Why Layering?



- No layering: each new application has to be **re-**implemented for every network technology!

Why Layering?

- Solution: introduce an intermediate layer that provides a **unique** abstraction for various network technologies



Layering

- Advantages
 - Modularity – protocols easier to manage and maintain
 - Abstract functionality –lower layers can be changed **without** affecting the upper layers
 - Reuse – upper layers can reuse the functionality provided by lower layers
- Disadvantages
 - Layering introduces overhead – less efficient

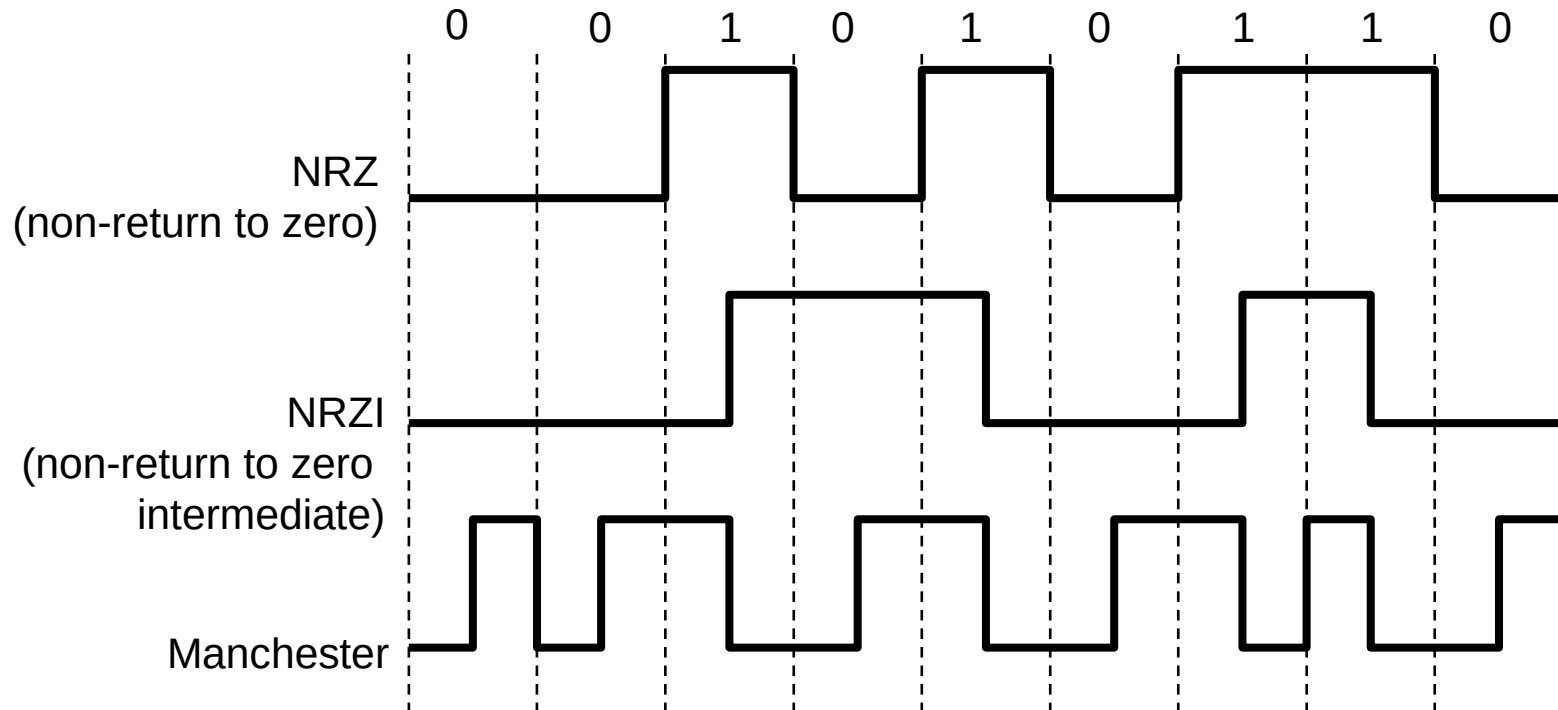
OSI Model Concepts

- Service – says **what** a layer does
- Interface – says **how** to **access** the service
- Protocol – says **how** is the service **implemented**
 - a set of rules and formats that govern the communication between two peers

Physical Layer (1)

- Service: move the information between two systems connected by a physical link
- Interface: specifies how to send a bit
- Protocol: coding scheme used to represent a bit, voltage levels, duration of a bit
- Examples: specification of coaxial cable, optical fiber links; transmitters, receivers; form factor

Encoding Schemes



Datalink Layer (2)

- Services that can be included:
 - framing, i.e., attach frame separator
 - send data frames between peers
 - other optional:
 - arbitrate the access to common physical media
 - ensure reliable transmission
 - provide flow control
- Interface: send a data unit (packet) to a machine connected to **same** physical media
- Protocol: physical layer addresses (e.g., Ethernet address), implement Medium Access Control (MAC) (e.g., CSMA/CD)...

Marking the Beginning/Ending of a Frame

- Use a special bit sequence
- Problem: what happens if this sequence appears in the data payload?
- Use **bit stuffing** technique; e.g., assume both beginning, ending bit sequences are 01111110
 - sender: inserts a 0 after 5 consecutive 1s
 - receiver: when it sees 5 1s makes decision on next **two** bits
 - if next bit 0 (this is a stuffed bit), remove it
 - if next bit 1, look at the next bit
 - ! if 0, receiver has seen 01111110
 - ! if 1 this is an error, discard the frame (receiver has seen 01111111)

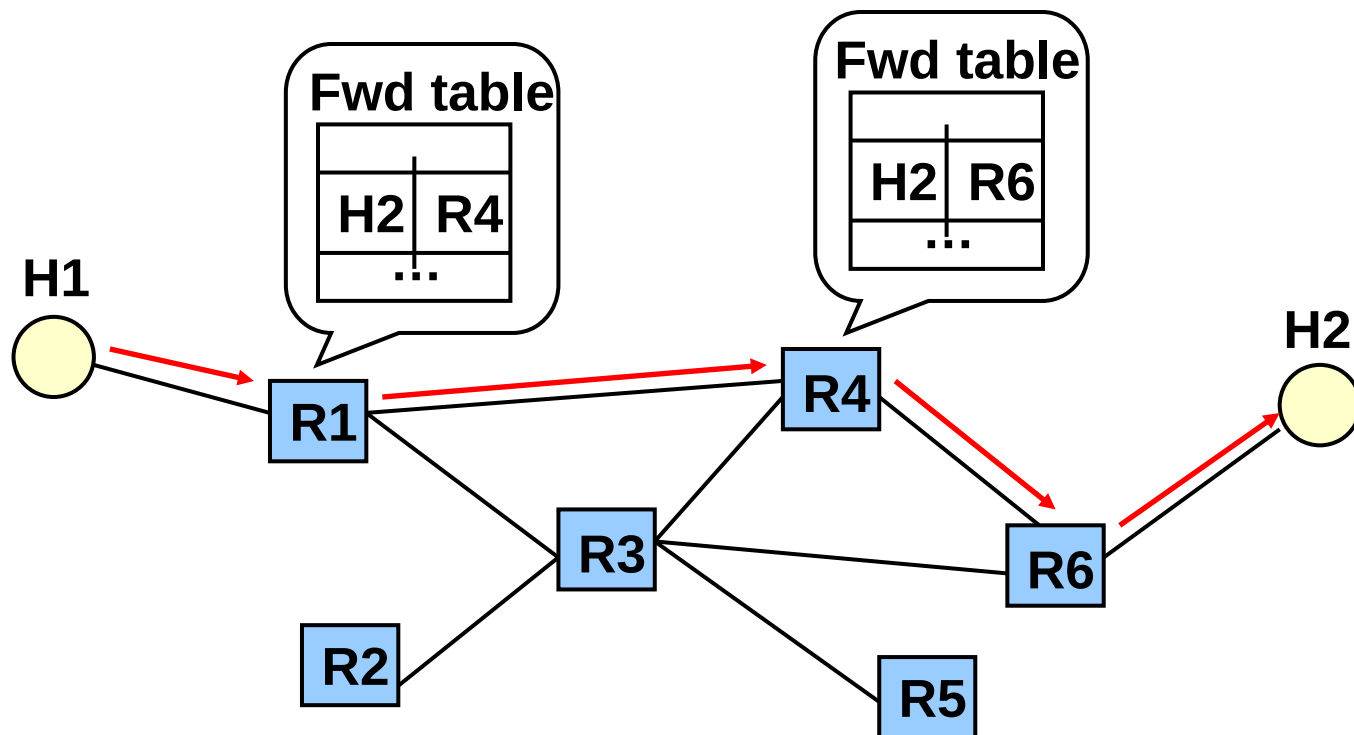
- Send: 1111110011
 - 11111010011
- Received: 11111 10011
- Send: 01111110
 - 011111010
- Received: 011111 10
- If received: 01111110 → special frame
- If received: 01111111 → error

Network Layer (3)

- Services that can be included:
 - routing
 - packet forwarding: forward a packet to specified destination
 - perform segmentation/reassembly if applicable
 - other optional:
 - packet scheduling
 - buffer management
- Interface: send a packet to a specified destination
- Protocol: define global unique addresses (e.g. IP address); construct routing tables, routing; packet forwarding

Example: Routing

- Data plane: use Forwarding Table to forward packets
- Control plane: construct and maintain Forwarding Tables (e.g., Distance Vector, Link State protocols)



Transport Layer (4)

- Services that can be included:
 - (sender side) multiplex multiple transport packet streams to network layer packet stream
 - (receiver side) split network layer packets into multiple transport packet streams
 - other optional: provide an **error-free** and **flow-controlled** end-to-end connection (e.g., TCP)
 - Interface: send a packet to specified destination
- Protocol: *may* implement reliability, flow control, congestion control
- Examples: TCP and UDP

Session Layer (5)

- Service:
 - manage different transport layer streams for the same application
 - synchronization, e.g., provide check points for long transfers
- Interface: depends on service
- Protocols: tie together different streams, insert checkpoints, implement roll-back functions
- Session layer is typically not used as a separate layer

Presentation Layer (6)

- Service:
 - Delivery and formatting of information to the application layer for further processing or display
 - Convert data between various representations
 - E.g., the conversion of an EBCDIC-coded text file to an ASCII-coded file
 - Encryption
- Interface: depends on service
- Protocol: define data formats, and rules to convert from one format to another
- In many widely used applications and protocols, no distinction is made between the presentation and application layers

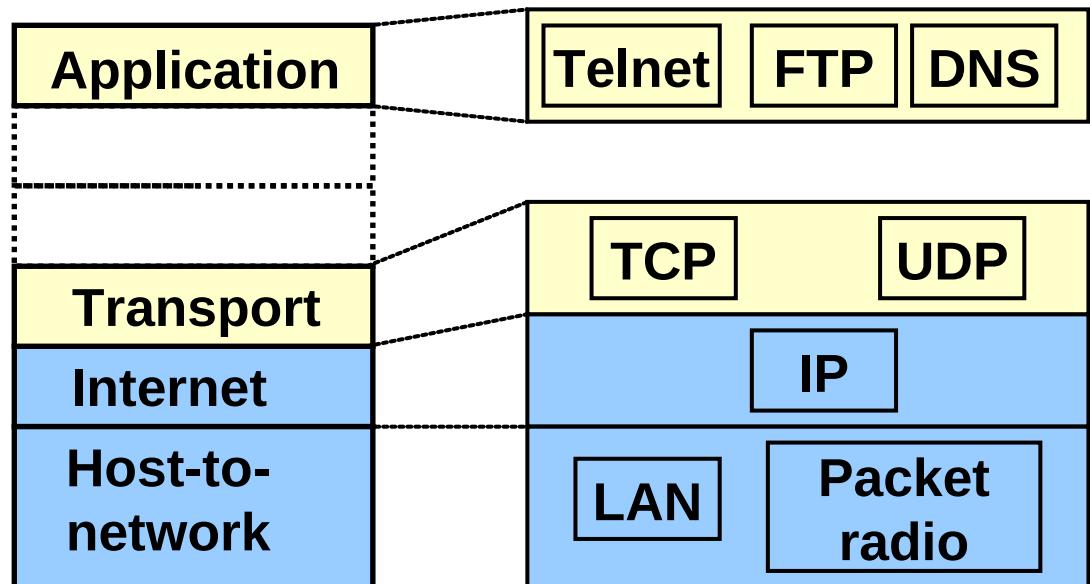
Application Layer (7)

- Service: any service provided to the end user
- Interface: depends on the application
- Protocol: depends on the application
- Examples: FTP, Telnet, WWW browser

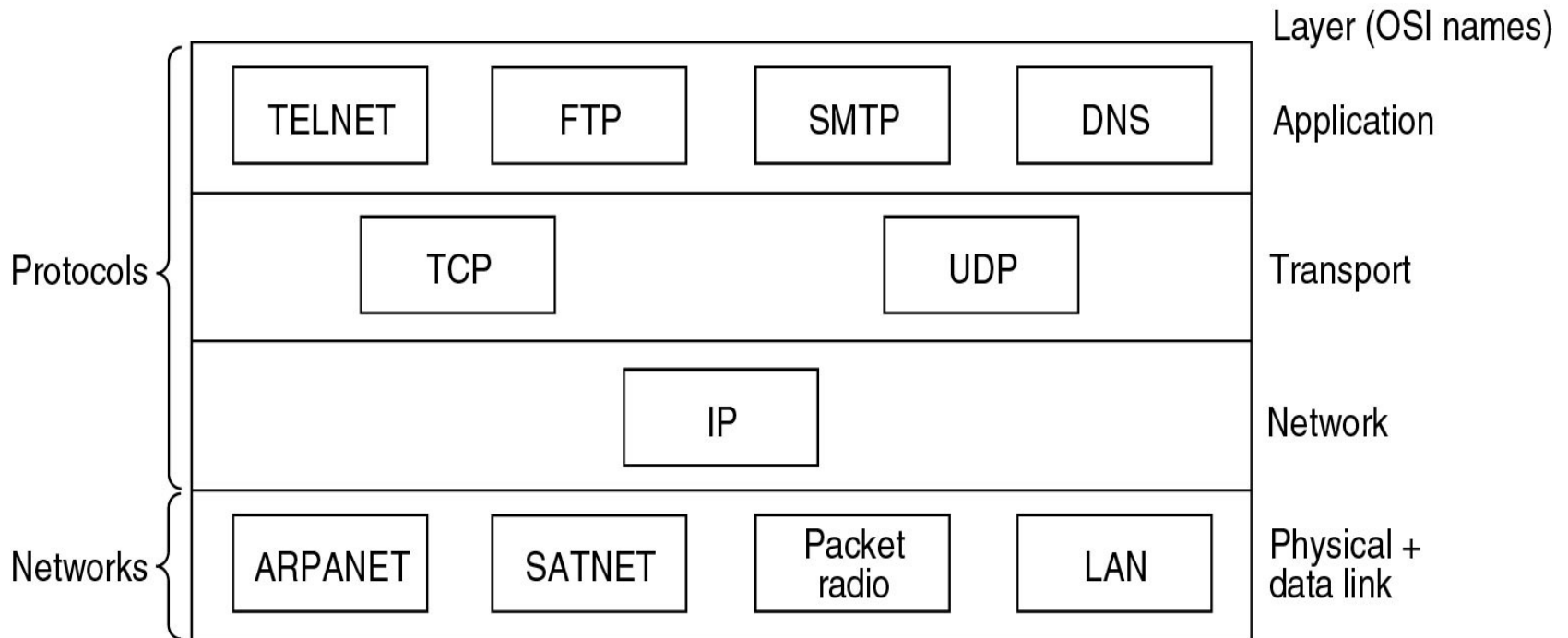
OSI vs. TCP/IP stack

- OSI: conceptually define: service, interface, protocol
- Internet: provide a successful implementation

Application
Presentation
Session
Transport
Network
Datalink
Physical

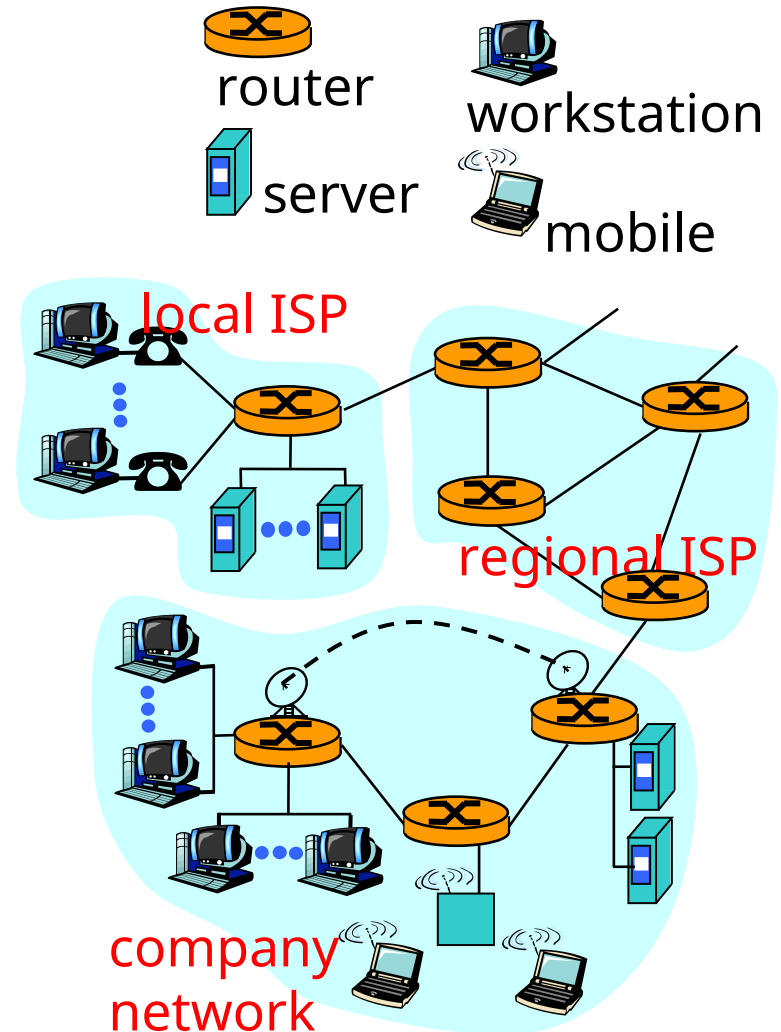


TCP/IP protocol stack



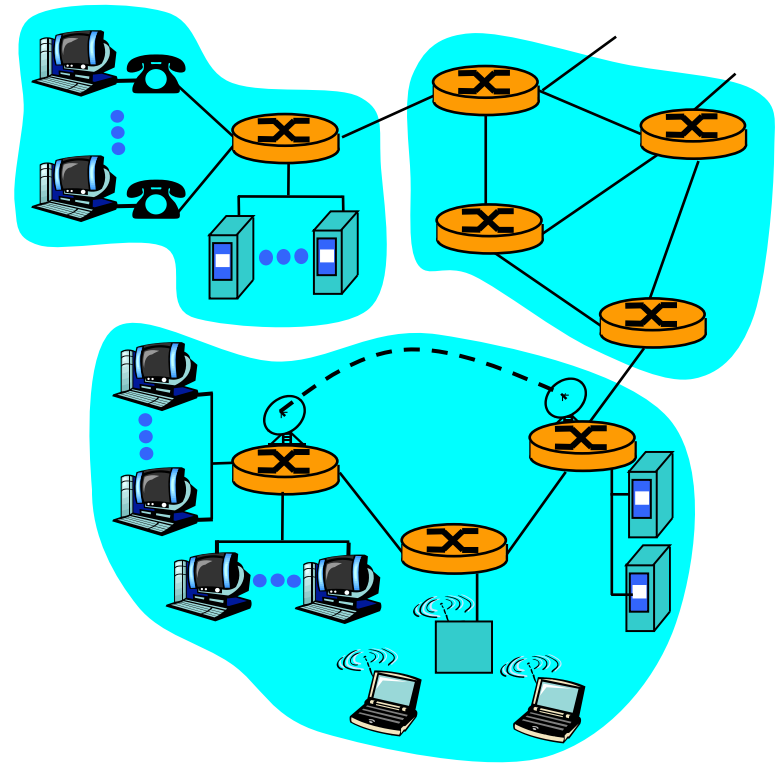
What's the Internet: “nuts and bolts” view

- *Internet*: “network of networks”
 - loosely hierarchical
- *protocols* control sending, receiving of msgs
 - e.g., TCP, IP, HTTP, FTP, PPP



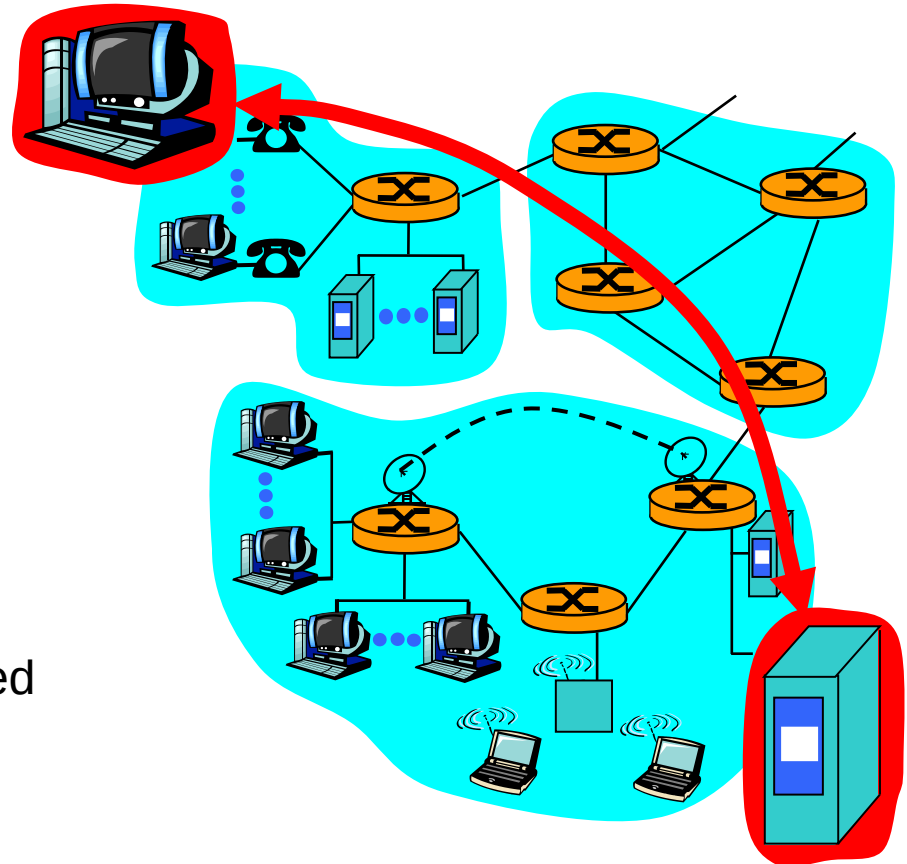
Network structure

- **Network edge:**
 - applications and hosts
 - **Access networks, physical media:** communication links
- **Network core:**
 - routers
 - network of networks



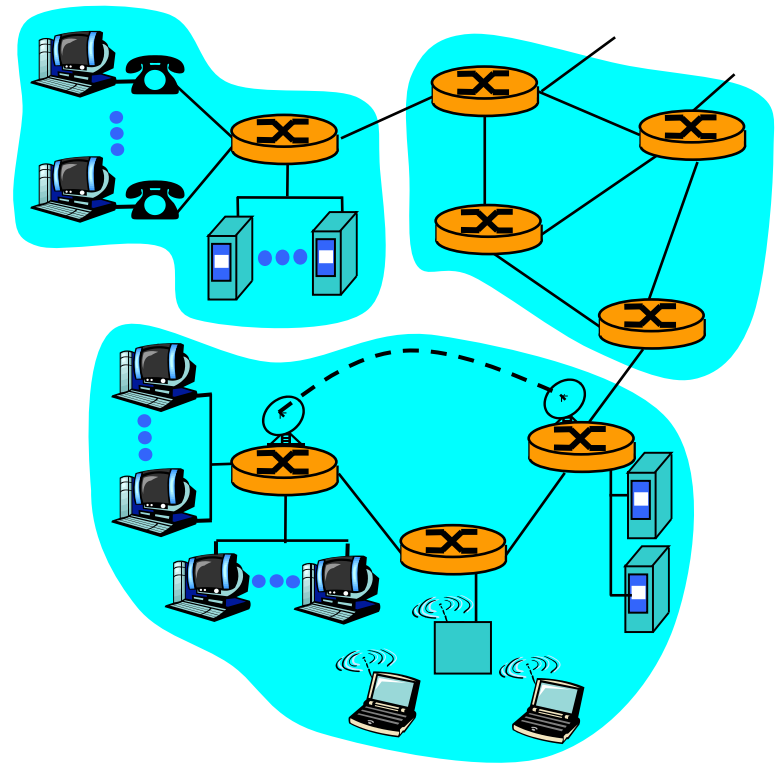
The network edge:

- **End systems (hosts):**
 - run application programs
 - e.g. Web, email
 - at “edge of network”
- **Client/server model**
 - client host requests, receives service from always-on server
 - e.g. Web browser/server; email client/server
- **Peer-peer model:**
 - minimal (or no) use of dedicated servers
 - e.g. Gnutella, KaZaA



What's the Internet: a service view

- **Communication *infrastructure*** enables distributed applications:
 - Web, email, games, e-commerce, database., voting, file (MP3) sharing
- **Communication services provided to apps:**
 - connectionless
 - connection-oriented



Network edge: connection-oriented service

Goal: data transfer between end systems

- *handshaking*: setup (prepare for) data transfer is needed ahead of time
 - Hello, hello back human protocol
 - *set up “state”* in two communicating hosts
 - E.g. TCP

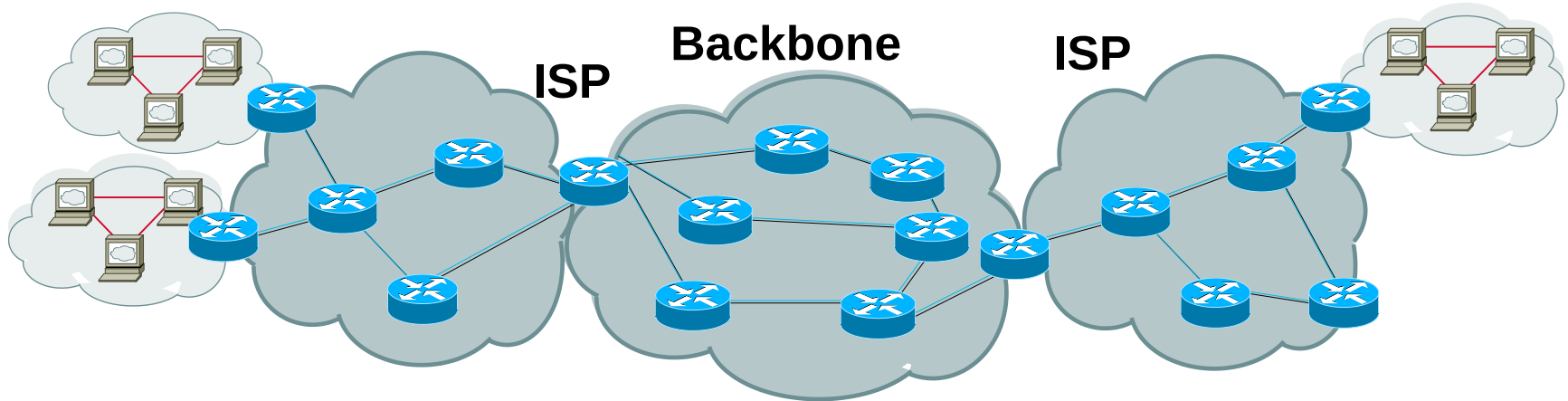
Network edge: connectionless service

Goal: data transfer between end systems

- same as before!
- No handshaking

Internet structure and ISPs

Internet Physical Infrastructure



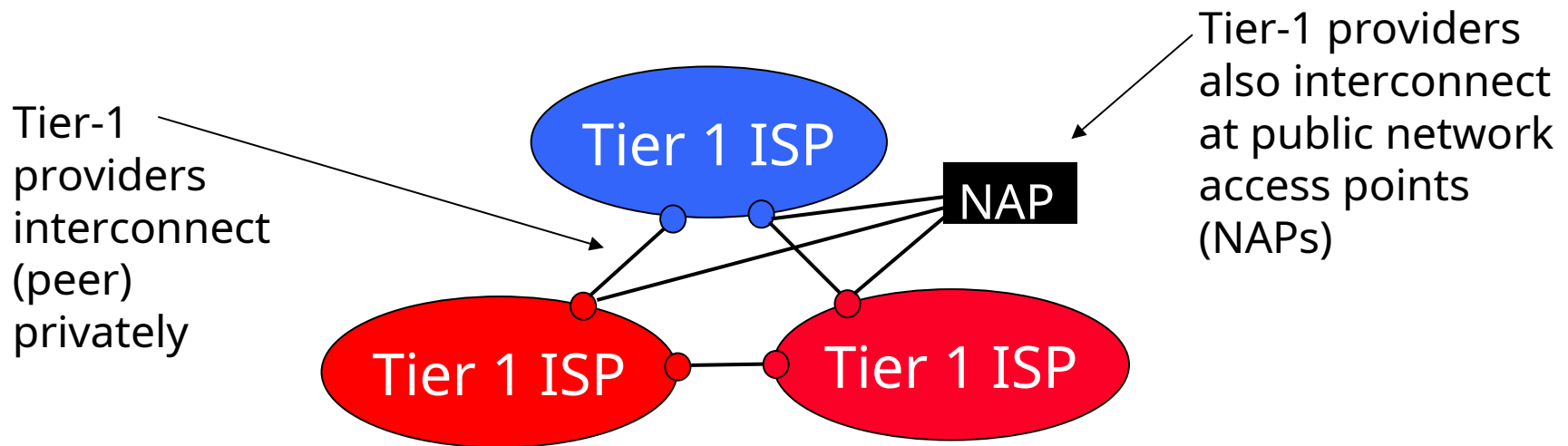
- Residential Access
 - Modem
 - DSL
 - Cable modem
 - Satellite
- Enterprise/ISP access, Backbone transmission
 - T1/T3, DS-1 DS-3
 - OC-3, OC-12
 - ATM vs. SONET, vs. WDM
- Campus network
 - Ethernet, ATM
- Internet Service Providers
 - access, regional, backbone
 - Network Access Point (NAP)

OARnet

- http://www.osc.edu/oarnet/about_oarnet/oarnetbackbone.gif
- [Third frontier network](#)

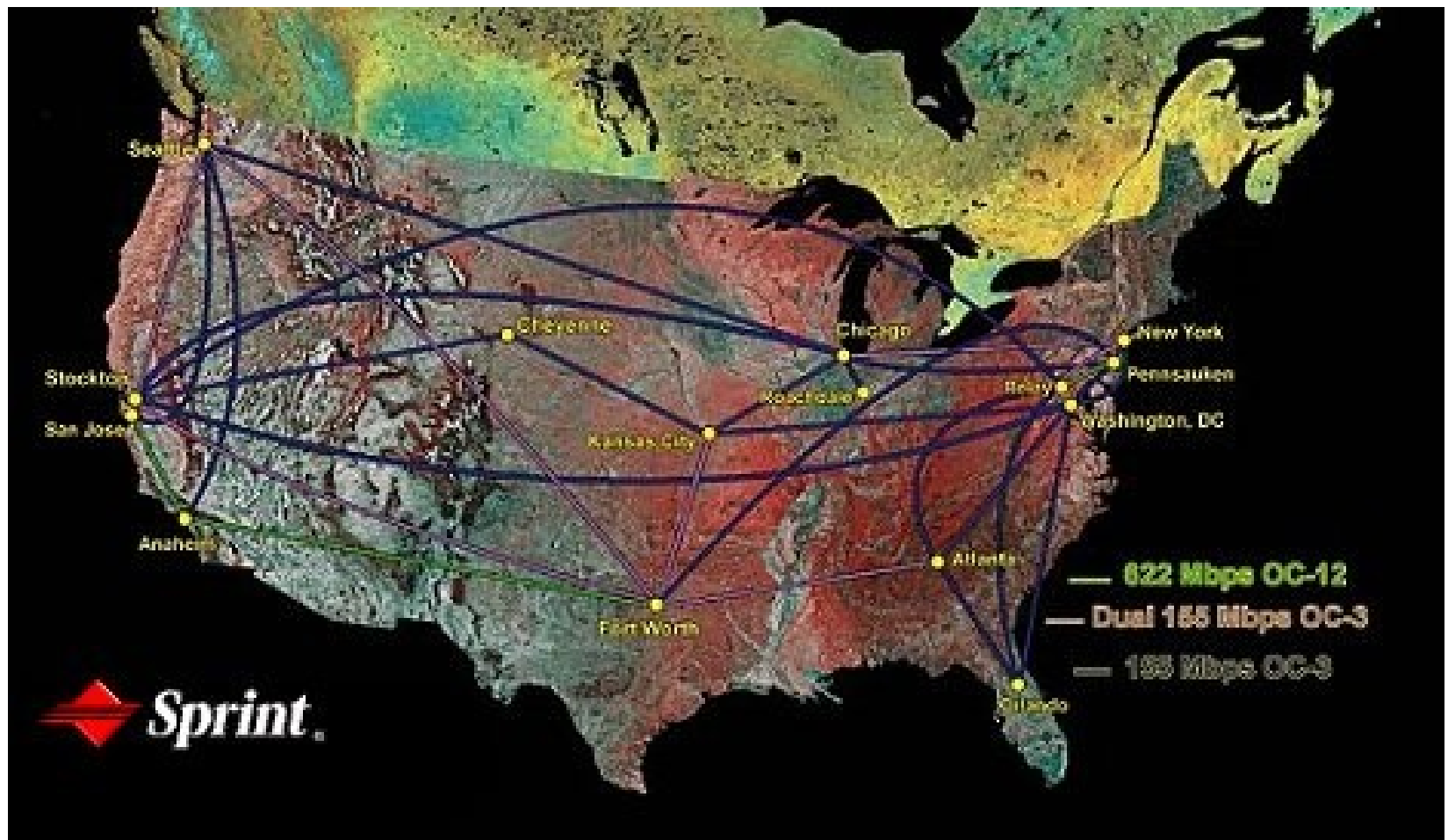
Internet structure: network of networks

- Roughly hierarchical
- **At center: “tier-1” ISPs** (e.g., UUNet, BBN/Genuity, Sprint, AT&T), national/international coverage
 - treat each other as equals



Tier-1 ISP: e.g., Sprint

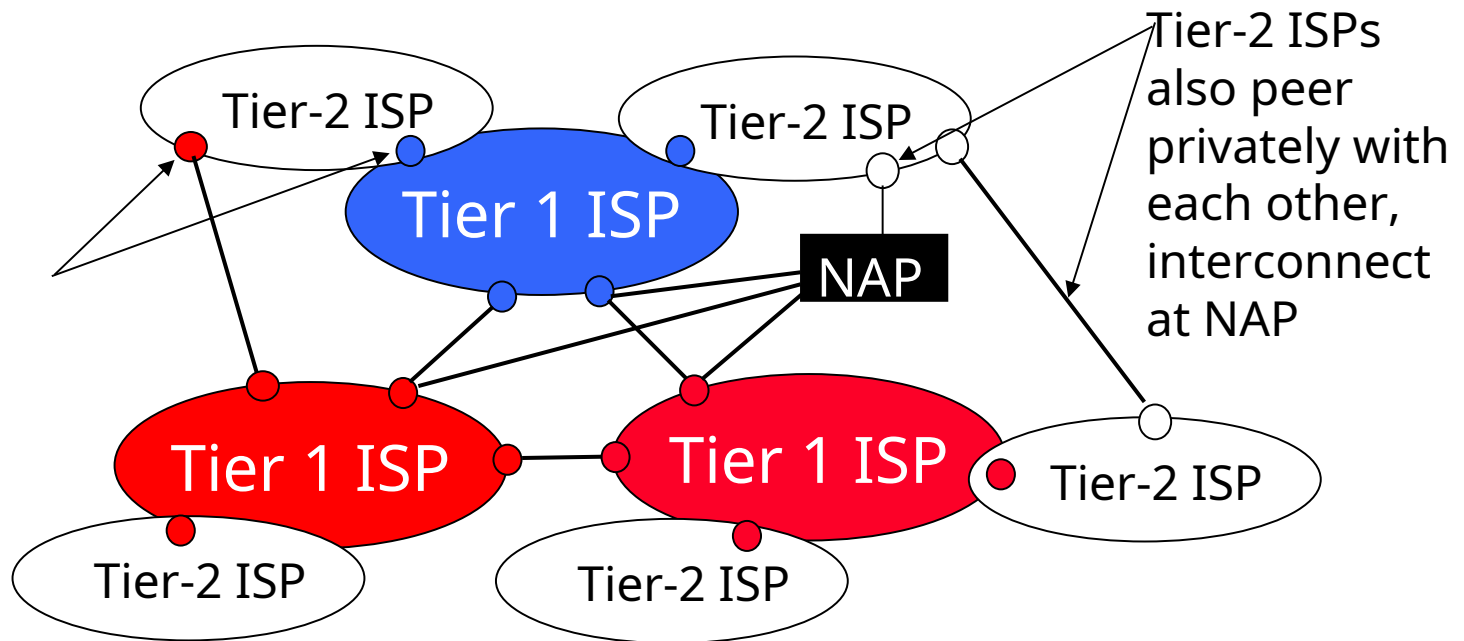
Sprint US backbone network



Internet structure: network of networks

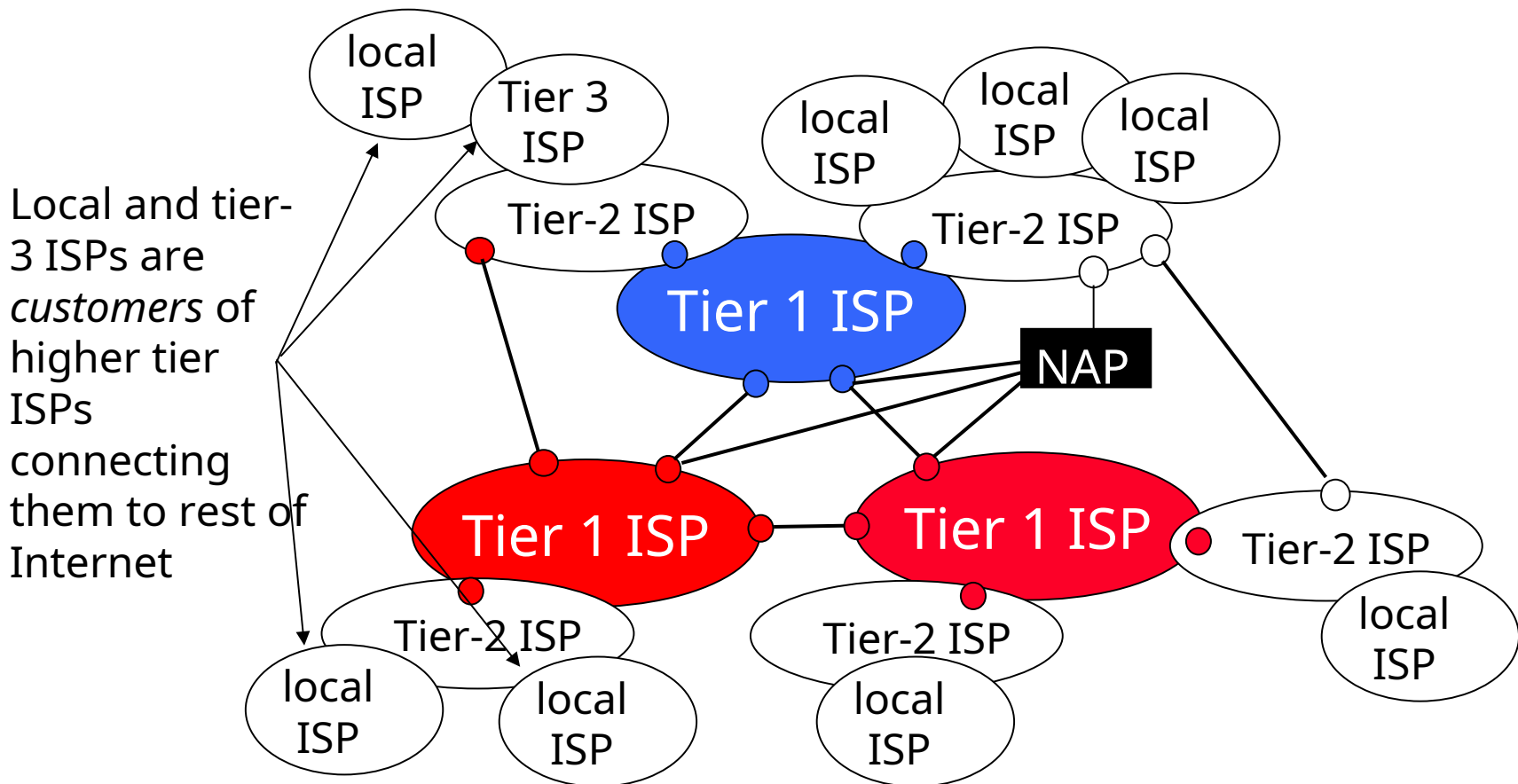
- “Tier-2” ISPs: smaller (often regional) ISPs
 - Connect to one or more tier-1 ISPs, possibly other tier-2 ISPs

Tier-2 ISP pays tier-1 ISP for connectivity to rest of Internet
□ tier-2 ISP is customer of tier-1 provider



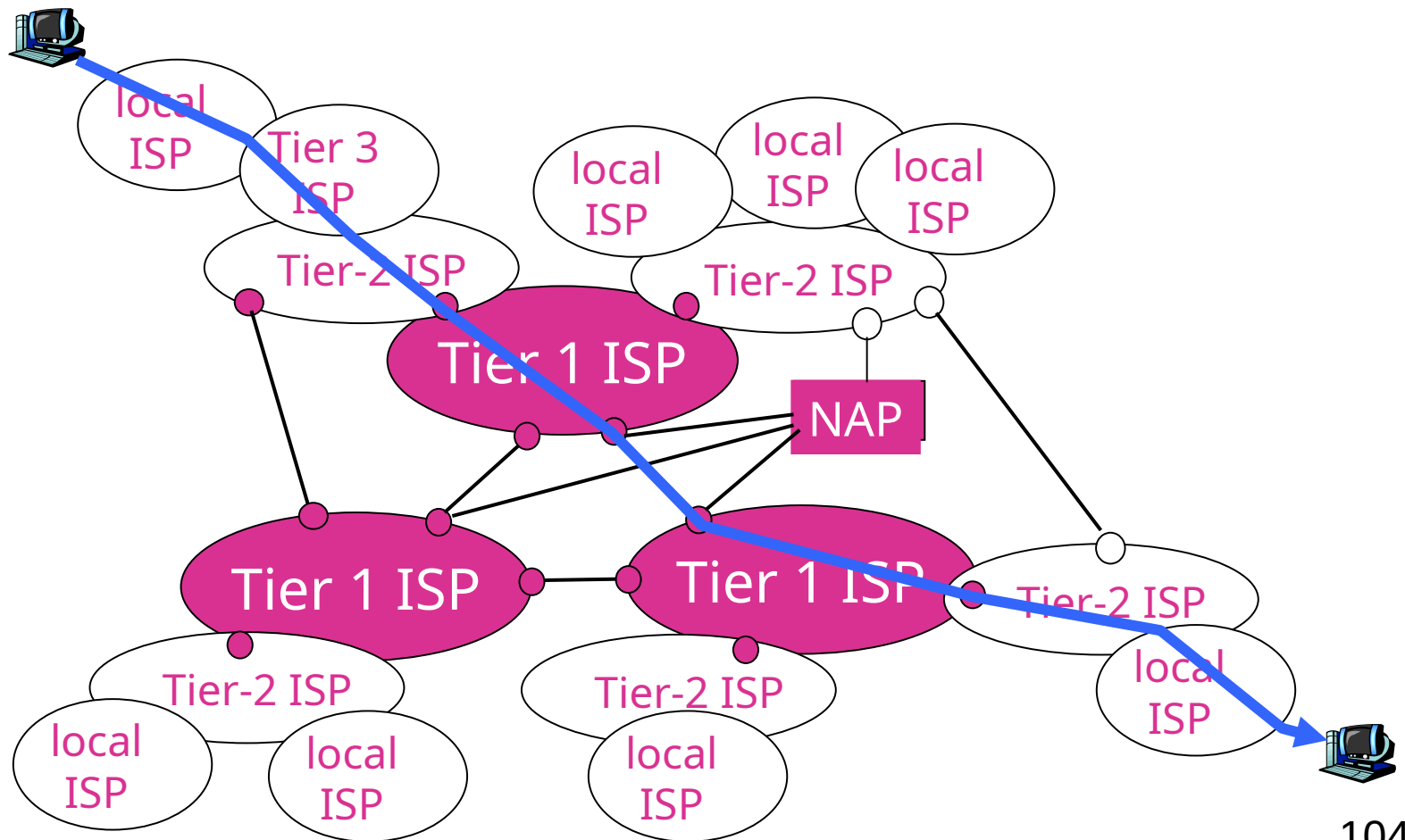
Internet structure: network of networks

- “Tier-3” ISPs and local ISPs
 - last hop (“access”) network (closest to end systems)



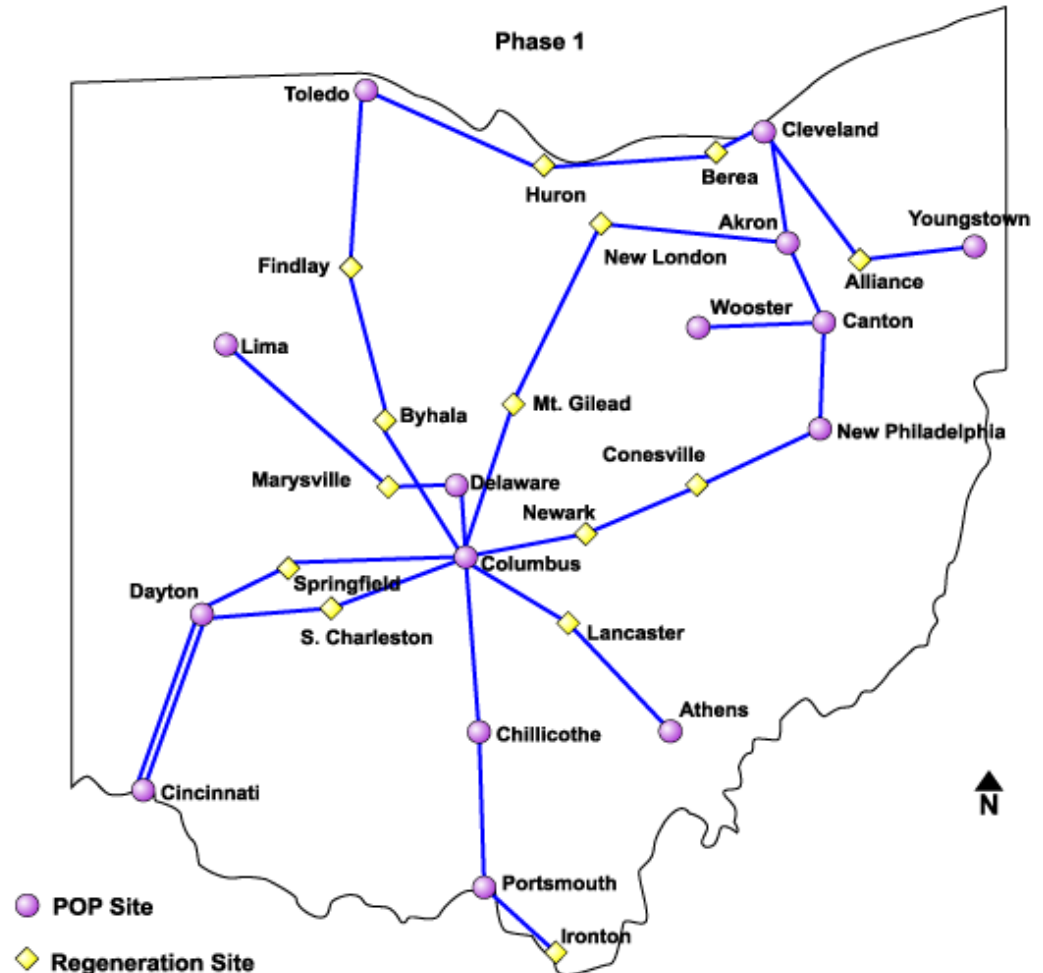
Internet structure: network of networks

- A packet passes through many networks!



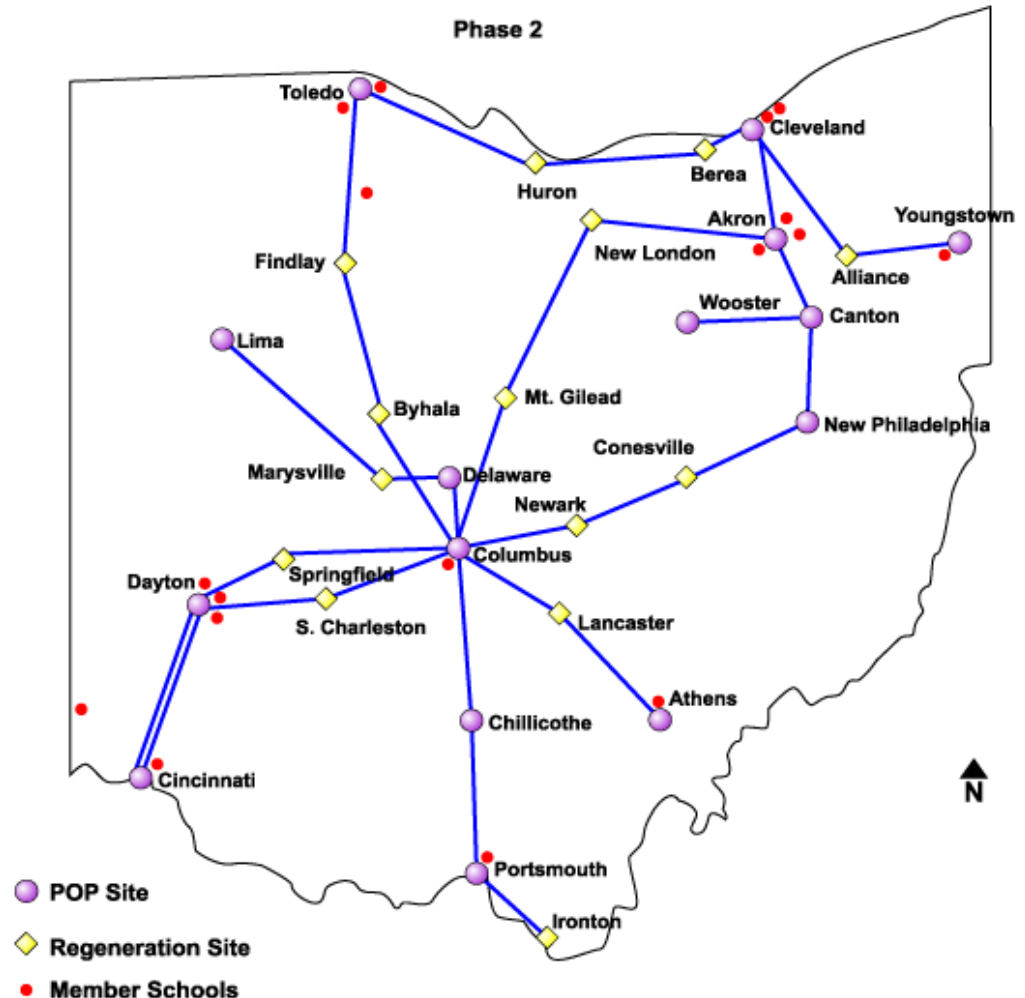
Statewide Core Backbone

- This backbone will connect 15 geographically dispersed hubs, or Points of Presence (POPs) throughout Ohio. Hub locations include: Akron, Athens, Canton, Chillicothe, Cincinnati, Cleveland, Columbus, Dayton, Delaware, Lima, New Philadelphia, Portsmouth, Toledo, Wooster, and Youngstown. More than a dozen regeneration sites will be inter-dispersed along the routes between the PoPs.



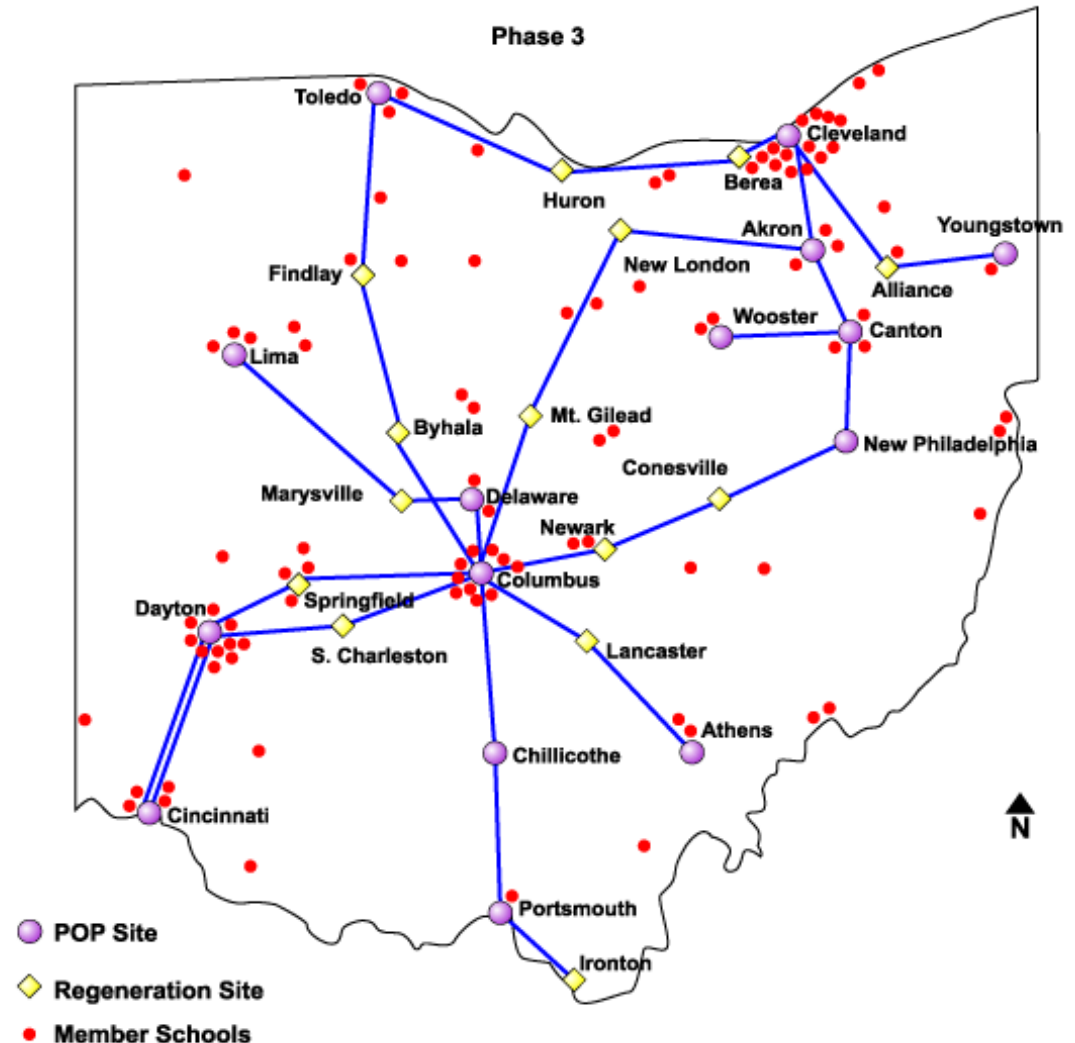
Last Mile Connectivity to 17 Major Higher Education Institutions

- Provide last mile fiber connectivity from the hubs described in Phase I to 17 individual campuses including: Bowling Green State University, Case Western Reserve University, Cleveland State University, Kent State University, Medical College of Ohio, Miami University, Northeastern Ohio Universities College of Medicine, The Ohio State University, Ohio University, University of Akron, University of Cincinnati, University of Dayton, University of Toledo, Wright State University, Central State University, Shawnee State University, Youngstown State University.



Last Mile Connectivity to Remaining Higher Education Institutions

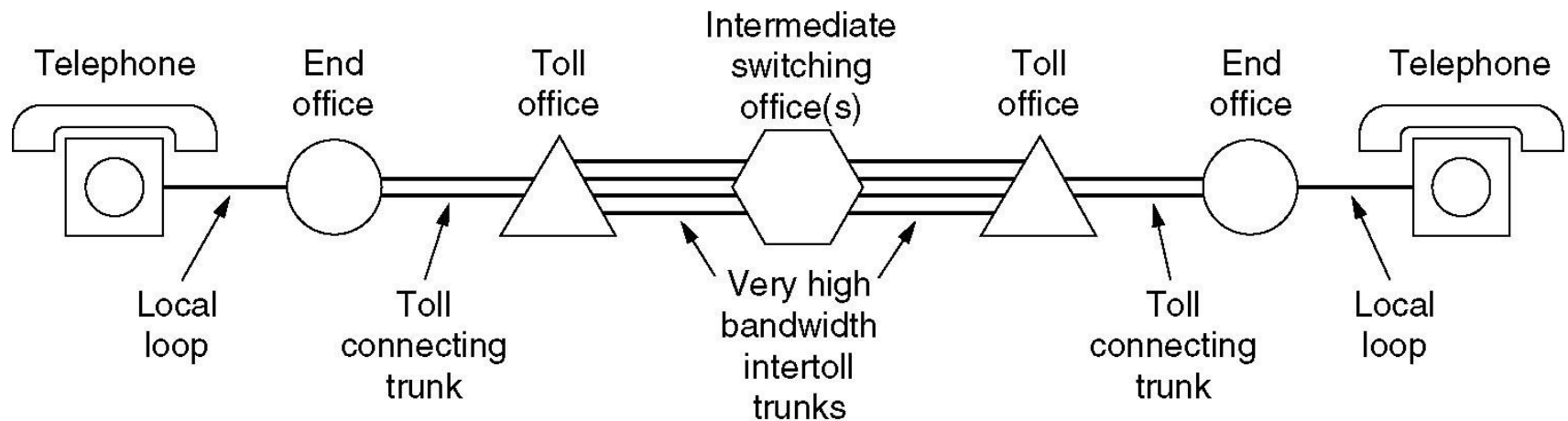
- Phase 3: provide last mile connectivity from the hubs, described in Phase I to more than 70 remaining higher education institutions in the state of Ohio.
- Phase 4: Connect Ohio's corporate research centers with the academic research community



Backup Slides

Structure of the Telephone System

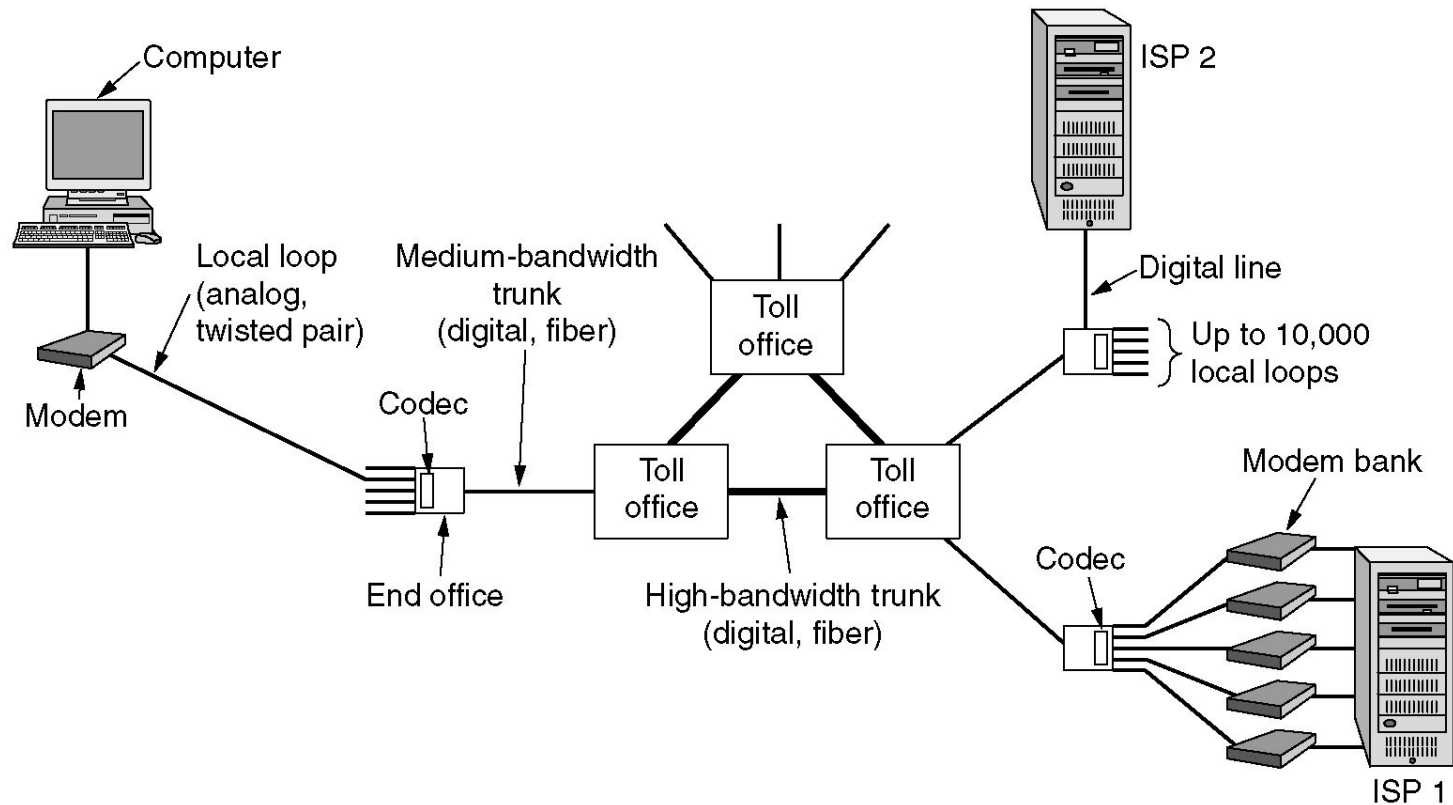
A typical circuit route for a medium-distance call.



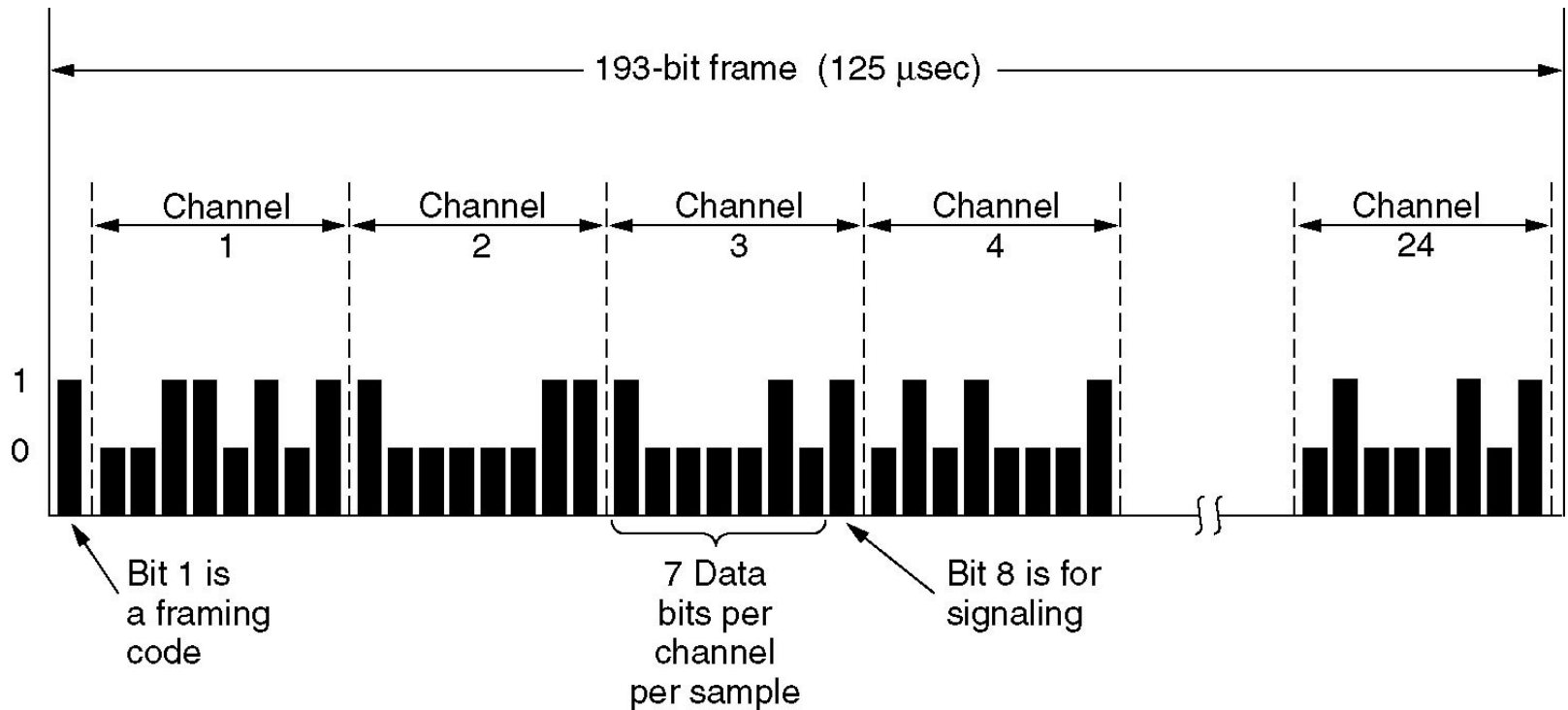
Major Components of the Telephone System

- Local loops
 - Analog twisted pairs going to houses and businesses
- Trunks
 - Digital fiber optics connecting the switching offices
- Switching offices
 - Where calls are moved from one trunk to another

The Local Loop: Modems, ADSL, and Wireless

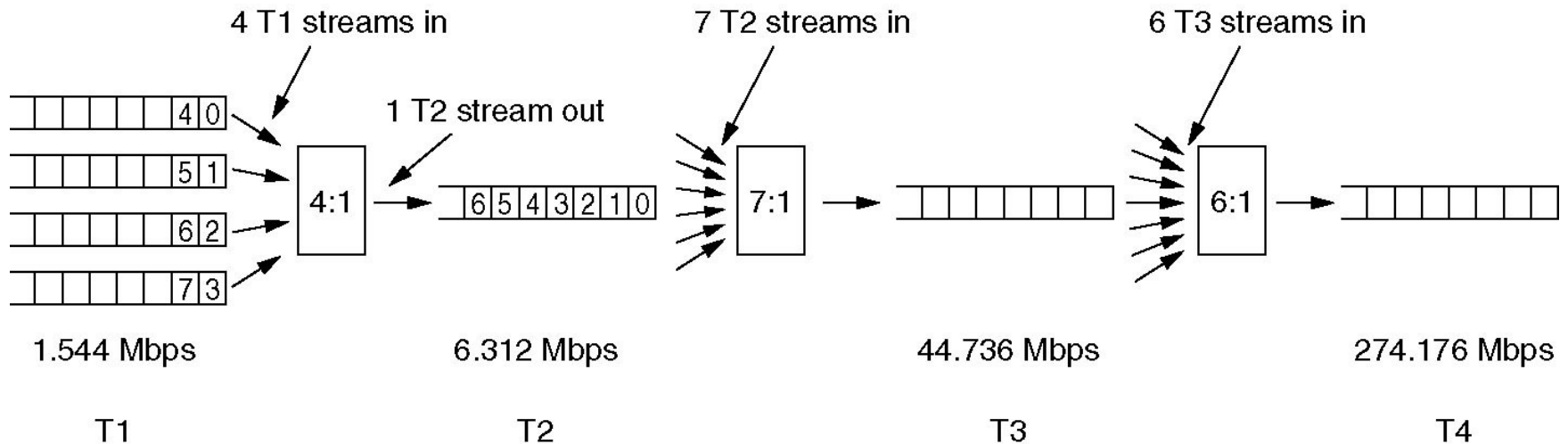


Time Division Multiplexing

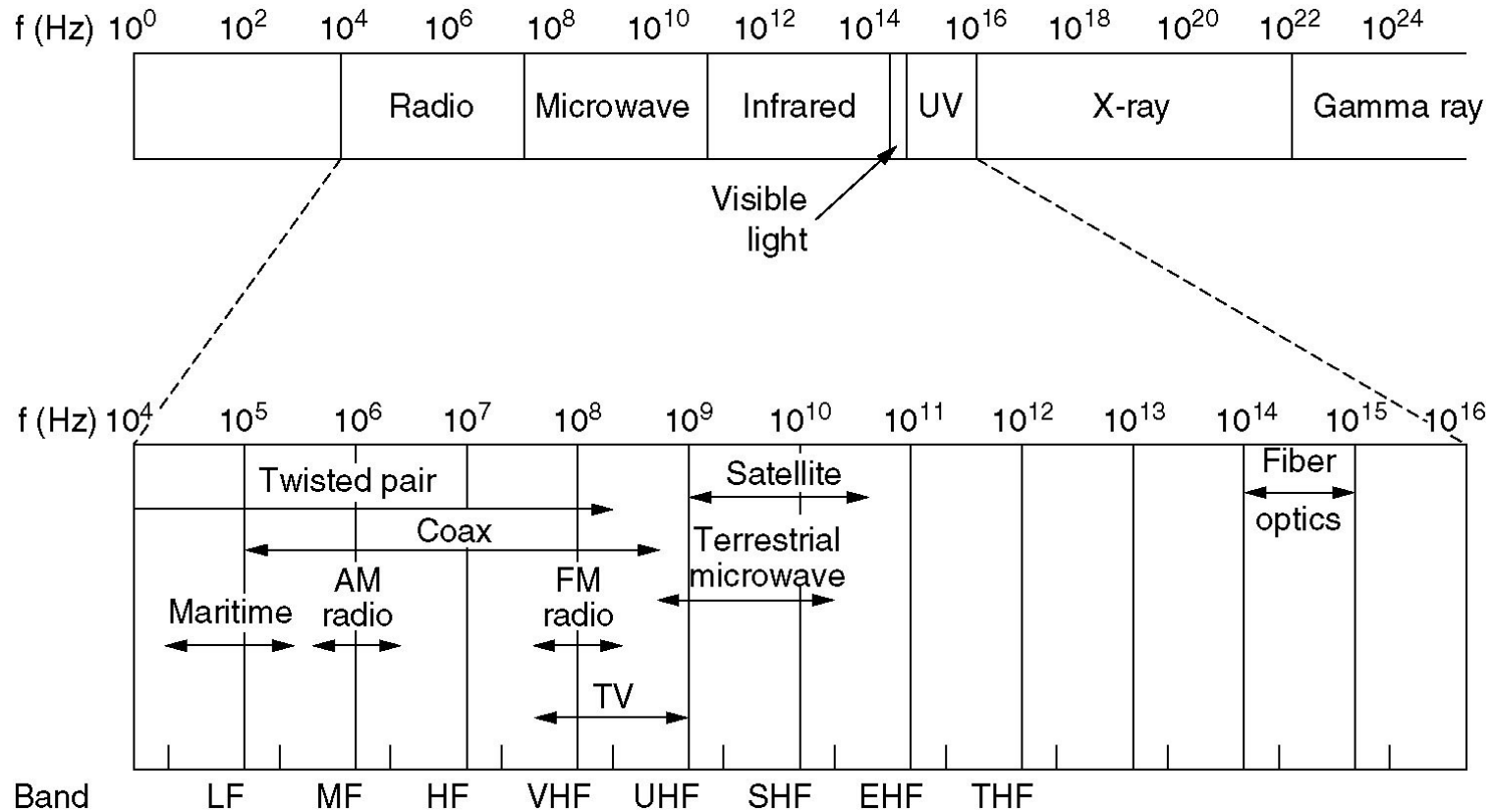


Time Division Multiplexing

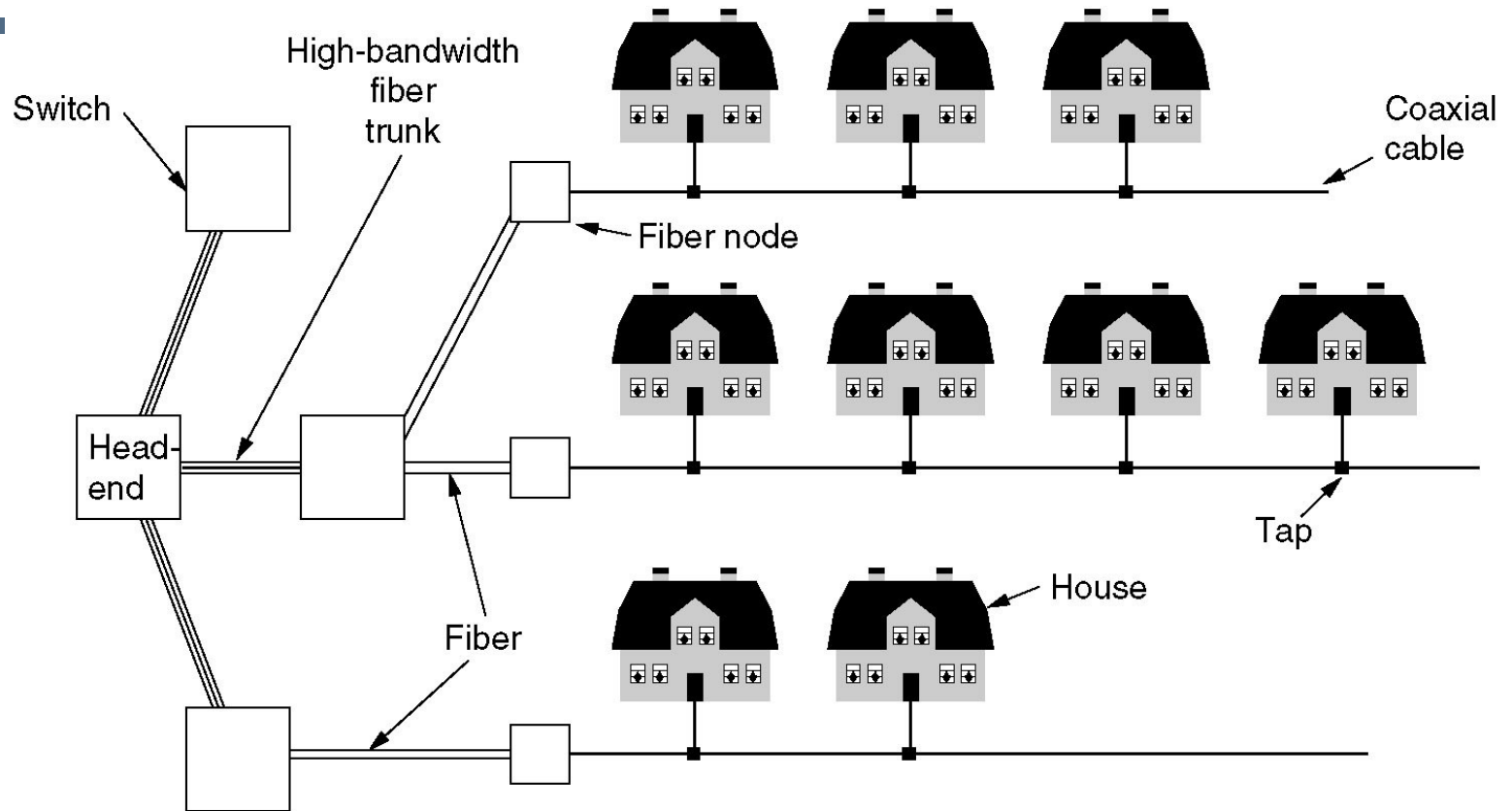
Multiplexing T1 streams into higher carriers.



The Electromagnetic Spectrum

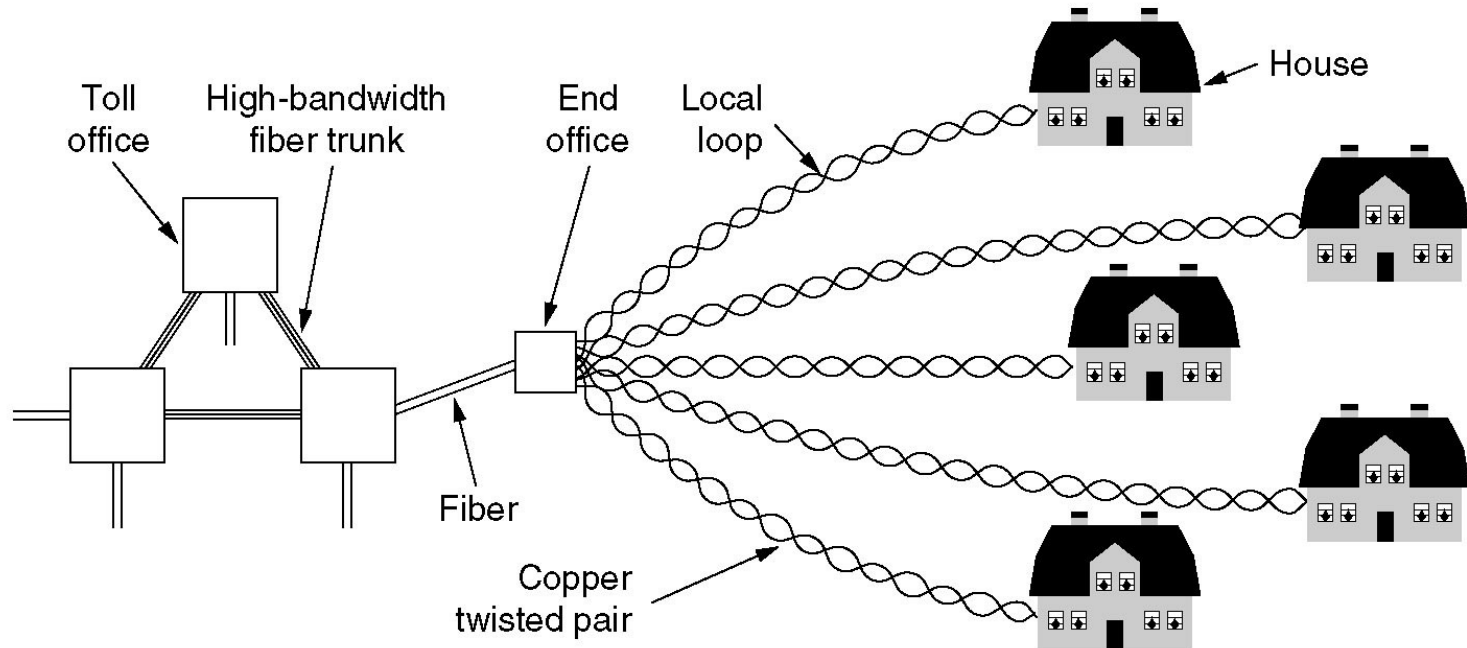


Internet over Cable



(a)

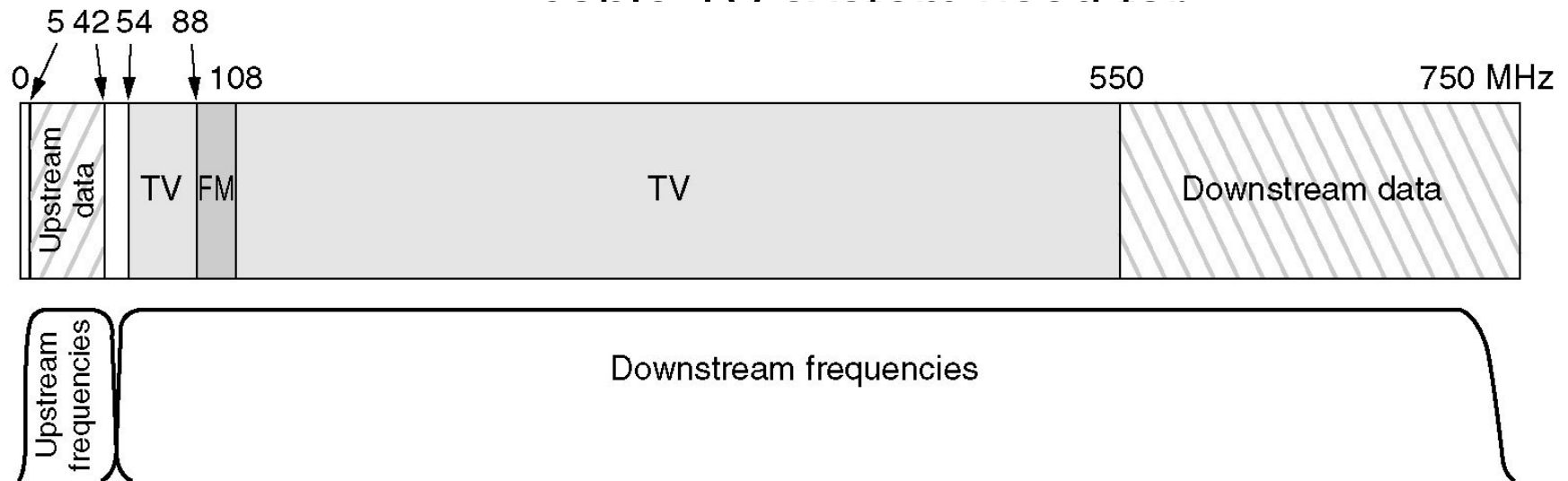
Internet over Cable (2)



(b)

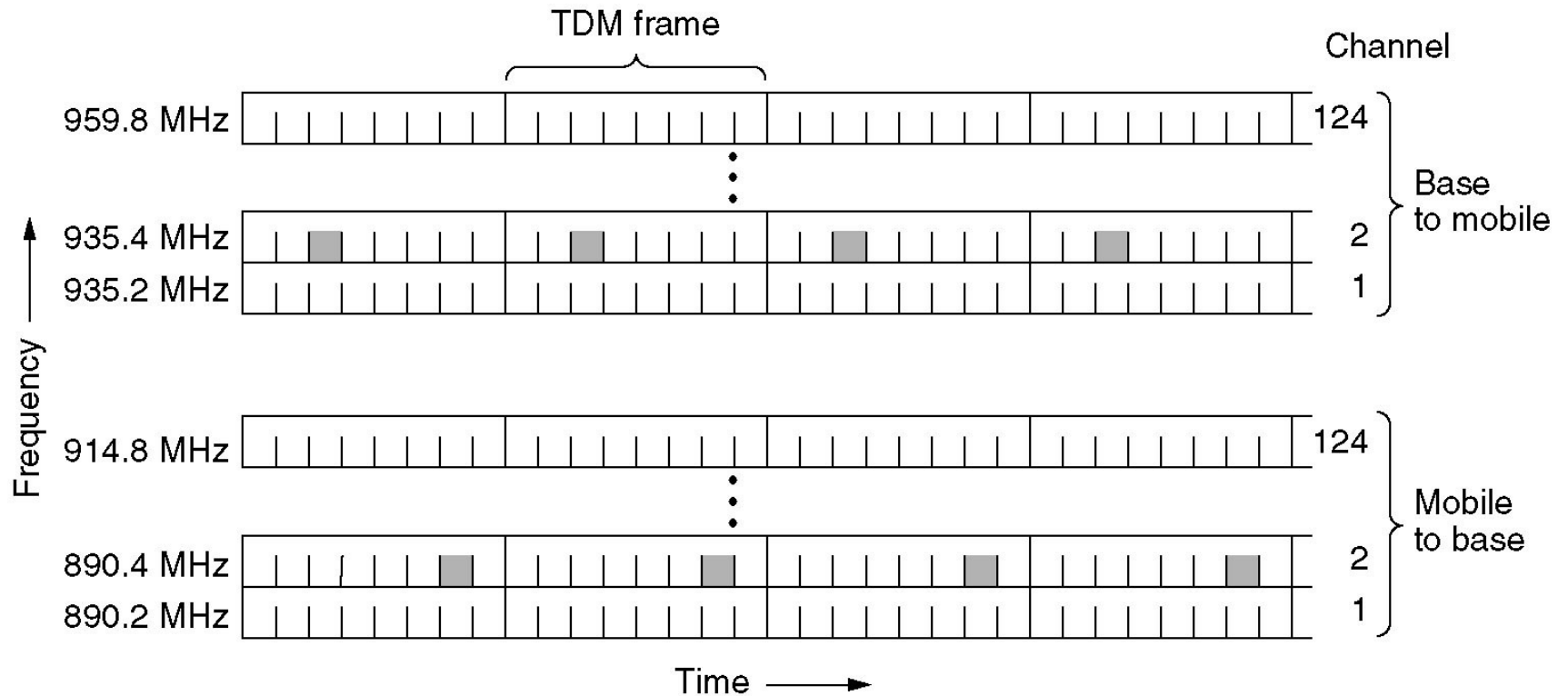
Spectrum Allocation

Frequency allocation in a typical



GSM

Global System for Mobile Communications



GSM (2)

